

SUBZERO+: EFFICIENT ZERO-TH-ORDER LLM FINE-TUNING VIA LARGE LEARNING RATES

Ziming Yu¹, Shuyao Xiao¹, Xingyu Zhao¹, Sike Wang¹, Pan Zhou², Peiyu Zang¹,
Xiangda Yan³, Yongjie Yang³, Jia Li^{1,4,5*}

¹ Beijing Normal University ² Singapore Management University ³ Xiaomi Inc.

⁴ Beijing Key Laboratory of Artificial Intelligence for Education

⁵ Engineering Research Center of Intelligent Technology and Educational Application (MOE)

{zimingyu, shuyaoxiao, xingyuzhao, sikewang}@mail.bnu.edu.cn
panzhou@smu.edu.sg, jiali@bnu.edu.cn

ABSTRACT

Zeroth-order (ZO) optimization enables backpropagation-free fine-tuning of large language models, but existing ZO methods suffer from high-variance gradient estimators, making convergence unstable and highly sensitive to learning rates. We propose SubZero+, an improved SubZero framework that improves stability in three complementary ways: (i) multi-query gradient estimation within layer-specific low-rank subspaces to reduce variance without exhibiting the multi-query paradox; (ii) a subspace Adam optimizer that performs adaptive updates using in-subspace multi-query gradient statistics; and (iii) a sign correction for QR-based subspace construction to ensure Haar-distributed projection matrices, eliminating implementation-dependent orientation ambiguity. Experiments on models from 1.3B to 32B across SuperGLUE, under both full-parameter tuning and LoRA, show that SubZero+ consistently outperforms prior ZO baselines, enlarges the stable learning-rate range, and narrows the gap to first-order methods with minimal extra memory overhead.

1 INTRODUCTION

Fine-tuning large language models (LLMs), such as OPT (Zhang et al., 2022), LLaMA (Touvron et al., 2023), GPT (Achiam et al., 2023), and Qwen (Yang et al., 2025), is a standard paradigm for adapting pre-trained models to downstream tasks. Most existing approaches rely on backpropagation (BP) to compute first-order (FO) gradients for optimization (Amari, 1993; Kingma & Ba, 2015). As a principled alternative, zeroth-order (ZO) optimization enables fine-tuning LLMs without backward passes (Malladi et al., 2023; Zhang et al., 2024; Liu et al., 2024; Yu et al., 2025). By eliminating BP, ZO methods can substantially reduce activation-memory overhead, simplify training pipelines, and make fine-tuning feasible in settings where BP is impractical or inefficient (Liu et al., 2026).

Despite the promise of ZO optimization (Malladi et al., 2023), matching the convergence behavior of FO methods remains challenging. A key bottleneck is the high variance of gradient estimates, which typically grows with the parameter dimension d (Duchi et al., 2015; Nesterov & Spokoiny, 2017). This becomes particularly severe for fine-tuning LLMs, where d is extremely large, often resulting in a significant performance gap relative to FO optimizers. Numerous efforts have focused on developing improved gradient estimators and variance-reduction strategies. Representative directions include imposing sparsity priors on perturbations (Liu et al., 2024; Guo et al., 2024), exploiting low-rank structure (Chen et al., 2025; Yu et al., 2025), incorporating second-order Hessian information (Zhao et al., 2025), and orthogonalizing perturbations in random subspaces (Lang et al., 2026).

Among these methods, SubZero (Yu et al., 2025) is a promising approach for fine-tuning LLMs. It projects gradients onto layer-specific low-dimensional subspaces, which reduces both the difficulty and the variance of gradient estimation. Since fine-tuning gradients often exhibit low-rank structure (Zhang et al., 2023; Malladi et al., 2023; Zhao et al., 2024; Hao et al., 2024), these subspace approximations can preserve the most informative optimization directions. As a result, the variance

*Corresponding Author

SUBZERO+: 通过大学习率进行高效的零阶大语言模型微调

余子明¹, 肖书尧¹, 赵星宇¹, 王思科¹, 周攀², 臧培宇¹, 阎祥达³, 杨永杰³, 李佳^{1,4,5*}

¹ 北京师范大学 ² 新加坡管理大学 ³ 小米公司 ⁴ 北京教育人工智能重点实验室 ⁵ 智能技

术与教育应用工程研究中心（教育部）{zimingyu, shuyaoxiao, xingyuzhao,

sikewang}@mail.bnu.edu.cn panzhou@smu.edu.sg, jiali@bnu.edu.cn

摘要

零阶（ZO）优化使大型语言模型的无反向传播微调成为可能，但现有的 ZO 方法存在高方差梯度估计问题，导致收敛不稳定并对学习率高度敏感。我们提出了 SubZero+，一个改进的 SubZero 框架，通过三种互补方式提高稳定性：(i) 在层特定低秩子空间内进行多查询梯度估计，以在不出现多查询悖论的情况下降低方差；(ii) 子空间 Adam 优化器，利用子空间内的多查询梯度统计执行自适应更新；(iii) 基于 QR 的子空间构建签名修正，以确保 Haar 分布的投影矩阵，消除实现依赖的方向歧义。在对 1.3B 到 32B 模型进行的 SuperGLUE 实验中，无论是全参数微调还是 LoRA，SubZero+ 都始终优于之前的 ZO 基线，扩大了稳定学习率范围，并缩小了与 f 的差距。

1 引言

微调大型语言模型（LLMs），例如 OPT (Zhang 等, 2022)、LLaMA (Touvron 等, 2023)、GPT (Achiam 等, 2023) 和 Qwen (Yang 等, 2025)，是将预训练模型适应下游任务的标准范式。大多数现有方法依赖反向传播 (BP) 来计算用于优化的一阶 (FO) 梯度 (Amari, 1993; Kingma & Ba, 2015)。作为一个有原则的替代方案，零阶 (ZO) 优化能够在不进行反向传播的情况下微调 LLM (Malladi 等, 2023; Zhang 等, 2024; Liu 等, 2024; Yu 等, 2025)。通过消除 BP，ZO 方法可以显著降低激活内存开销，简化训练流程，并使微调在 BP 不切实际或效率低下的情况下成为可能。

刘等, 2026年。

尽管 ZO 优化 (Malladi 等, 2023) 有一定的前景，但要匹配 FO 方法的收敛行为仍然具有挑战性。一个关键的瓶颈是梯度估计的高方差，这通常会随着参数维度的增加而增长 (Duchi 等, 2015; Nesterov & Spokoiny, 2017)。对于微调大型语言模型（LLMs）来说，这一问题尤为严重，其中参数维度 d 极其庞大，通常导致与 FO 优化器相比出现显著的性能差距。大量研究工作集中于开发改进的梯度估计方法和方差降低策略。典型方向包括对扰动施加稀疏先验 (Liu 等, 2024; Guo 等, 2024)、利用低秩结构 (Chen 等, 2025; Yu 等, 2025)、引入二阶 Hessian 信息 (Zhao 等, 2025) 以及在随机子空间中对扰动进行正交化 (Lang 等, 2026)。

在这些方法中，SubZero (Yu 等, 2025) 是一种用于微调大型语言模型（LLM）的有前途的方法。它将梯度投影到特定层的低维子空间，从而降低了梯度估计的难度和方差。由于微调梯度通常表现出低秩结构 (Zhang 等, 2023; Malladi 等, 2023; Zhao 等, 2024; Hao 等, 2024)，这些子空间近似可以保留最有信息量的优化方向。因此，方差

*通讯作者

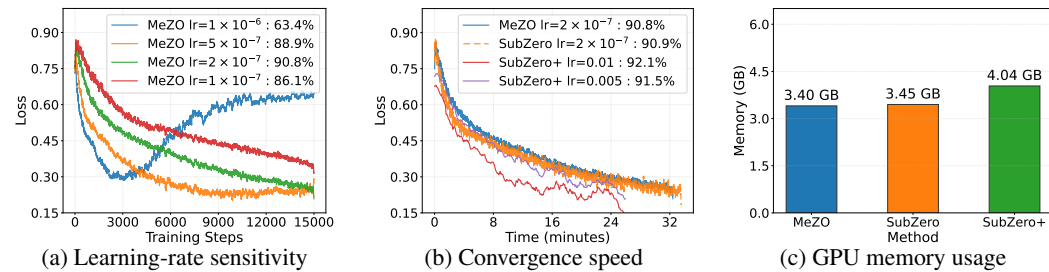


Figure 1: Learning-rate sensitivity, convergence speed, and GPU memory usage for full-parameter tuning of OPT-1.3B on SST-2. (a) MeZO is highly sensitive to the learning rate. (b) SubZero+ produces smoother training curves and supports larger learning rates, leading to faster convergence. (c) SubZero+ adds only ~ 0.6 GB of GPU memory overhead compared with MeZO and SubZero while improving both stability and performance.

reduction from operating in a smaller space can outweigh the loss in gradient fidelity, yielding substantial practical improvements. However, like many ZO fine-tuning methods, SubZero is sensitive to hyperparameters, especially the learning rate. A large learning rate amplifies the effect of estimator noise, which can cause oscillations or divergence (Ghadimi & Lan, 2013). As shown in Figure 1(a), MeZO (Malladi et al., 2023) requires an unusually small learning rate to remain stable. We observe that this sensitivity also limits SubZero in practice. While averaging multiple queries can mitigate noise, it increases query cost. Moreover, the gains from additional queries are often sublinear and may even suffer from the multi-query paradox (Lin et al., 2025). Therefore, enabling larger effective learning rates for SubZero through further variance reduction remains an open challenge.

To achieve faster convergence with larger learning rates, we propose SubZero+, which improves training stability in three complementary aspects. First, we perform multi-query gradient estimation within the same low-rank subspace to reduce variance while mitigating the multi-query paradox. Second, because multiple queries yield a more accurate gradient estimate, we apply Adam (Kingma & Ba, 2015) for parameter updates inside the subspace. This leads to more effective training of LLMs (Glentis et al., 2026) while retaining a low memory footprint. Third, we introduce a sign correction for the QR-based subspace construction to ensure that the resulting projection matrices are Haar-distributed (Mezzadri, 2007), removing implementation-dependent orientation ambiguities.

Our main contributions are summarized as follows:

- We extend SubZero’s single-query estimator by averaging multiple independent perturbations within the same low-rank subspace. This reduces gradient variance without incurring the multi-query paradox. All statistics are computed in-subspace, so the memory footprint remains comparable to inference-time (see Figures 1(c) and 2).
- We propose a low-rank subspace Adam for parameter updates. Adam is executed entirely within the low-rank subspace, and multi-query gradient norms are used to set per-layer, per-direction adaptive step sizes. This adds overhead proportional to the subspace size and yields low-noise, curvature-aware updates that improve stability and optimization quality.
- We apply a sign correction to the QR-based subspace construction so that the projection matrices are Haar-distributed, eliminating orientation ambiguities introduced by implementation details.
- We conduct extensive experiments on models from 1.3B to 32B across SuperGLUE, under both full-parameter tuning (FT) and LoRA (Hu et al., 2022). SubZero+ consistently outperforms prior ZO baselines, significantly enlarges the stable learning-rate range (see Figure 1(b)), and narrows the performance gap to first-order methods.

2 RELATED WORK

ZO Gradient Estimation. ZO optimization replaces BP with finite-difference gradient estimates using only forward passes. MeZO (Malladi et al., 2023) pioneered SPSA-based ZO fine-tuning for LLMs (Spall, 1992), achieving inference-level memory but suffering from variance that scales with parameter dimension d (Nesterov & Spokoiny, 2017; Duchi et al., 2015). Subsequent work

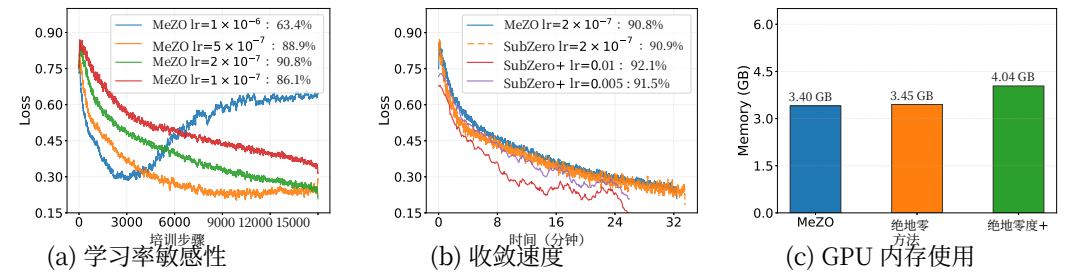


图 1: OPT-1.3B 在 SST-2 上进行全参数调优时的学习率敏感性、收敛速度和 GPU 内存使用情况。 (a) MeZO 对学习率高度敏感。 (b) SubZero+ 产生更平滑的训练曲线，并支持更大的学习率，从而实现更快的收敛。 (c) 与 MeZO 和 SubZero 相比，SubZero+ 仅增加 ~ 0.6 GB 的 GPU 内存开销，同时提升了稳定性和性能。

在较小空间中操作的减少可能超过梯度保真度的损失，从而产生显著的实际改进。然而，与许多零次优化 (ZO) 微调方法一样，SubZero 对超参数非常敏感，尤其是学习率。较大的学习率会放大估计器噪声的影响，这可能导致振荡或发散 (Ghadimi & Lan, 2013)。如图 1(a) 所示，MeZO (Malladi 等, 2023) 需要异常小的学习率才能保持稳定。我们观察到，这种敏感性在实际中也限制了 SubZero。虽然对多个查询取平均可以减轻噪声，但会增加查询成本。此外，额外查询带来的收益通常是次线性的，甚至可能受到多查询悖论的影响 (Lin 等, 2025)。因此，通过进一步降低方差来使 SubZero 能够使用更大的有效学习率仍然是一个未解决的挑战。

为了在更大的学习率下实现更快的收敛，我们提出了 SubZero+，在三个互补方面提高了训练的稳定性。首先，我们在相同的低秩子空间内执行多查询梯度估计，以在减小多查询悖论的同时降低方差。其次，由于多次查询会产生更准确的梯度估计，我们在子空间内对参数更新应用 Adam (Kingma & Ba, 2015)。这使得 LLM (Glentis 等, 2026) 的训练更加高效，同时保持低内存占用。第三，我们为基于 QR 的子空间构建引入了符号校正，以确保生成的投影矩阵是 Haar 分布的 (Mezzadri, 2007)，消除了实现依赖的方向模糊性。

我们的主要贡献总结如下：

- 我们通过在同一低秩子空间内对多个独立扰动取平均，扩展了 SubZero 的单查询估计器。这在不引发多查询悖论的情况下减少了梯度方差。所有统计数据都在子空间内计算，因此内存占用与推理时相当 (见图 1(c) 和图 2)。
- 我们提出了一种用于参数更新的低秩子空间 Adam 方法。Adam 完全在低秩子空间内执行，并使用多查询梯度范数来设置每层、每个方向的自适应步长。这增加了与子空间大小成比例的开销，并产生低噪声、考虑曲率的更新，从而提高了稳定性和优化质量。
- 我们对基于 QR 的子空间构造应用符号校正，以使投影矩阵服从 Haar 分布，从而消除由实现细节引入的方向歧义。
- 我们在从 1.3B 到 32B 的模型上，在 SuperGLUE 上进行了广泛的实验，涵盖全参数调优 (FT) 和 LoRA (Hu 等, 2022)。+ SubZero 始终优于先前的零阶基线，显著扩大了稳定学习率范围 (见图 1(b))，并缩小了与一阶方法的性能差距。

2 相关工作

零阶梯度估计。零阶优化用仅需前向传播的有限差分梯度估计替代了反向传播。MeZO (Malladi 等, 2023) 开创了基于 SPSA 的大语言模型零阶微调 (Spall, 1992)，实现了推理级别的内存占用，但存在随参数维度增加的方差问题 d (Nesterov & Spokoiny, 2017; Duchi 等, 2015)。随后工作

mitigates this through sparse perturbations (Liu et al., 2024; Guo et al., 2024) and low-rank subspace projections (Chen et al., 2025; Yu et al., 2025). LOZO (Chen et al., 2025) factorizes perturbations as low-rank matrices, while SubZero (Yu et al., 2025) enforces column-orthonormal projection matrices via QR decomposition. Both operate with a single query per iteration, limiting their variance reduction capability.

ZO Parameter Update. Standard ZO methods including MeZO (Malladi et al., 2023) and SubZero (Yu et al., 2025) apply vanilla SGD to the estimated gradients. LOZO (Chen et al., 2025) applies SGDM within the low-rank subspace, incurring negligible memory overhead. TeZO (Sun et al., 2025) further exploits temporal low-rankness across gradients, and can be extended to an Adam variant. However, the Adam extension does not yield consistent improvement over its SGD counterpart. Indeed, MeZO, SubZero, and (Guo et al., 2025) all observe that Adam offers no advantage over SGD in ZO fine-tuning, as the noisy gradient estimates render the second-moment preconditioning unreliable. Extending adaptive optimizers to ZO is challenging for two reasons: the high variance of gradient estimates makes the second-moment estimate unreliable, and full-size moment buffers violate the memory-efficient premise. Running adaptive optimization within a compressed subspace, where moment buffers are proportionally reduced, remains an open challenge.

Multi-Query ZO. Averaging $K > 1$ independent perturbation evaluations reduces per-step gradient variance by a factor of K (Duchi et al., 2015). However, Lin et al. (2025) identified the *multi-query paradox*: under a fixed budget of total forward passes, using K queries reduces the number of training steps by a factor of K , and the per-step variance reduction is exactly canceled by the fewer update steps, yielding no net gain over single-query training. Recent works seek to break this trade-off through gradient normalization (Dang et al., 2026), projection alignment (Lin et al., 2025), and gradient orthogonalization (Liu et al., 2025). A complementary direction is to leverage shared structure across queries, such as a common low-rank subspace, to align signal components before averaging, potentially circumventing the paradox without additional computational overhead.

3 PRELIMINARIES

We first introduce the standard ZO optimization formulation and the SubZero framework upon which SubZero+ builds. We adopt the notation convention from SubZero (Yu et al., 2025): bold lower-case letters ($\mathbf{w}$) denote vectors, bold upper-case letters ($\mathbf{W}$) denote matrices, and calligraphic letters ($\mathcal{W}$) denote sets of matrices.

Problem Setup. Consider fine-tuning a pre-trained LLM consisting of l layers, where the trainable parameters of the i -th layer are represented as a matrix $\mathbf{W}_i \in \mathbb{R}^{m_i \times n_i}$ (e.g., weight matrices of linear projections in attention and feed-forward blocks). Let $\mathcal{W} = \{\mathbf{W}_i\}_{i=1}^l$ denote the set of all trainable parameter matrices, and let $\mathbf{w} = [\text{vec}(\mathbf{W}_1)^\top, \dots, \text{vec}(\mathbf{W}_l)^\top]^\top \in \mathbb{R}^d$ be their flattened concatenation, where $d = \sum_{i=1}^l m_i n_i$ is the total parameter count. The fine-tuning objective is to minimize a loss function $\mathcal{L}(\mathcal{W})$ over a training dataset $\mathcal{D}$.

ZO Gradient Estimation (MeZO). The classical ZO approach, popularized for LLM fine-tuning by MeZO (Malladi et al., 2023), employs the simultaneous perturbation stochastic approximation (SPSA) estimator (Spall, 1992). For a minibatch $\mathcal{B} \subset \mathcal{D}$, the central-difference ZO gradient estimate w.r.t. the flattened parameters $\mathbf{w}$ is:

$$\hat{\nabla}_{\mathbf{w}} \mathcal{L}(\mathbf{w}; \mathcal{B}) = \frac{\mathcal{L}(\mathbf{w} + \varepsilon \mathbf{z}; \mathcal{B}) - \mathcal{L}(\mathbf{w} - \varepsilon \mathbf{z}; \mathcal{B})}{2\varepsilon} \mathbf{z}, \quad (1)$$

where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ is a random Gaussian perturbation vector, and $\varepsilon > 0$ is the perturbation scale. This estimator requires only two forward passes and provides an unbiased estimate of the gradient of the smoothed objective $\mathbb{E}_{\mathbf{z}}[\mathcal{L}(\mathbf{w} + \varepsilon \mathbf{z})]$. However, its variance scales as $O(d)$ (Nesterov & Spokoiny, 2017), making gradient estimates extremely noisy for large d .

Subspace ZO Optimization (SubZero). SubZero (Yu et al., 2025) reduces gradient estimation variance by restricting perturbations to a low-rank subspace constructed *per layer*. For the i -th layer, SubZero generates two column-orthonormal projection matrices $\mathbf{U}_i \in \mathbb{R}^{m_i \times r}$ and $\mathbf{V}_i \in \mathbb{R}^{n_i \times r}$ via QR decomposition of random Gaussian matrices, where $r \ll \min\{m_i, n_i\}$ is the subspace rank. The perturbation for layer i is then constructed as:

$$\tilde{\mathbf{Z}}_i = \mathbf{U}_i \mathbf{Z}_i \mathbf{V}_i^\top \in \mathbb{R}^{m_i \times n_i}, \quad (2)$$

预印本。

通过稀疏扰动 (Liu 等, 2024; Guo 等, 2024) 和低秩子空间投影 (Chen 等, 2025; Yu 等, 2025) 来缓解这一问题。LOZO (Chen 等, 2025) 将扰动分解为低秩矩阵, 而 SubZero (Yu 等, 2025) 通过 QR 分解强制列正交归一投影矩阵。两者每次迭代都只使用一次查询, 限制了它们的方差降低能力。

ZO 参数更新。标准的 ZO 方法包括 MeZO (Malladi 等, 2023) 和 Sub-Zero (Yu 等, 2025) 对估计的梯度应用普通的 SGD。LOZO (Chen 等, 2025) 在低秩子空间内应用带动量的 SGD (SGDM), 几乎不增加内存开销。TeZO (Sun 等, 2025) 进一步利用梯度之间的时间低秩特性, 并可以扩展到 Adam 变体。然而, Adam 扩展在其 SGD 对应方法上并未带来一致的改进。实际上, MeZO、SubZero 和 (Guo 等, 2025) 都观察到, 在 ZO 微调中 Adam 相比 SGD 没有优势, 因为噪声梯度估计使二阶矩预处理不可靠。将自适应优化器扩展到 ZO 存在两方面的挑战: 梯度估计的高方差使二阶矩估计不可靠, 而全尺寸动量缓冲区违背了内存高效的前提。在压缩子空间中运行自适应优化, 其中动量缓冲区按比例减少

多查询 ZO。对 $K > 1$ 个独立扰动评估取平均可以将每步梯度方差减少 K 倍 (Duchi 等人, 2015)。然而, Lin 等人 (2025) 发现了多查询悖论: 在固定的总前向传播预算下, 使用 K 个查询会将训练步数减少 K 倍, 而每步方差的降低恰好被较少的更新步数抵消, 从而相比单查询训练没有净收益。近期的工作尝试通过梯度归一化 (Dang 等人, 2026)、投影对齐 (Lin 等人, 2025) 和梯度正交化 (Liu 等人, 2025) 来打破这一权衡。一个互补的方向是利用查询间共享的结构, 例如共同的低秩子空间, 在平均之前对信号分量进行对齐, 从而可能在不增加额外计算开销的情况下规避该悖论。

3 初步工作

我们首先介绍标准的零阶优化 (ZO) 公式以及 SubZero 框架, 这是 SubZero+ 基于的框架。我们采用 SubZero (Yu 等, 2025) 中的符号约定: 加粗的小写字母 ($\mathbf{w}$) 表示向量, 加粗的大写字母 ($\mathbf{W}$) 表示矩阵, 花体字母 ($\mathcal{W}$) 表示矩阵集合。

问题设置。考虑对一个由 l 层组成的预训练大语言模型进行微调, 其中第 i 层的可训练参数表示为矩阵 $\mathbf{W}_i \in \mathbb{R}^{m_i \times n_i}$ (例如, 注意力和前馈模块中线性投影的权重矩阵)。令 $\mathcal{W} = \{\mathbf{W}_i\}_{i=1}^l$ 表示所有可训练参数矩阵的集合, 并令 $\mathbf{w} = [\text{vec}(\mathbf{W}_1)^\top, \dots, \text{vec}(\mathbf{W}_l)^\top]^\top \in \mathbb{R}^d$ 表示它们的展平拼接, 其中 $d = \sum_{i=1}^l m_i n_i$ 是参数总数。微调的目标是在训练数据集 $\mathcal{D}$ 上最小化损失函数 $\mathcal{L}(\mathcal{W})$ 。

ZO 梯度估计 (MeZO)。经典的 ZO 方法, 由 MeZO (Malladi 等, 2023) 推广用于 LLM 微调, 采用了同时扰动随机逼近 (SPSA) 估计器 (Spall, 1992)。对于一个大批量 $\mathcal{B} \subset \mathcal{D}$, 相对于展平参数 $\mathbf{w}$ 的中心差分 ZO 梯度估计为:

$$\hat{\nabla}_{\mathbf{w}} \mathcal{L}(\mathbf{w}; \mathcal{B}) = \frac{\mathcal{L}(\mathbf{w} + \varepsilon \mathbf{z}; \mathcal{B}) - \mathcal{L}(\mathbf{w} - \varepsilon \mathbf{z}; \mathcal{B})}{2\varepsilon} \mathbf{z}, \quad (1)$$

其中 $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ 是一个随机高斯扰动向量, $\varepsilon > 0$ 是扰动尺度。该估计器只需要两次前向传递, 并提供平滑目标 $\mathbb{E}_{\mathbf{z}}[\mathcal{L}(\mathbf{w} + \varepsilon \mathbf{z})]$ 的梯度的无偏估计。然而, 其方差随 $O(d)$ 变化 (Nesterov & Spokoiny, 2017), 使得在大 d 时梯度估计极其噪声化。

子空间 ZO 优化 (SubZero)。SubZero (Yu 等, 2025) 通过将扰动限制在每层构建的低秩子空间中, 从而减少梯度估计的方差。对于第 i 层, SubZero 通过对随机高斯矩阵进行 QR 分解生成两个列正交投影矩阵 $\mathbf{U}_i \in \mathbb{R}^{m_i \times r}$ 和 $\mathbf{V}_i \in \mathbb{R}^{n_i \times r}$, 其中 $r \ll \min\{m_i, n_i\}$ 是子空间的秩。然后第 i 层的扰动构建为:

$$\tilde{\mathbf{Z}}_i = \mathbf{U}_i \mathbf{Z}_i \mathbf{V}_i^\top \in \mathbb{R}^{m_i \times n_i}, \quad (2)$$

where $\mathbf{Z}_i \in \mathbb{R}^{r \times r}$ is a low-dimensional random matrix with i.i.d. $\mathcal{N}(0, 1)$ entries. The loss difference is computed globally:

$$\rho = \frac{\mathcal{L}(\mathcal{W} + \varepsilon \tilde{\mathbf{Z}}; \mathcal{B}) - \mathcal{L}(\mathcal{W} - \varepsilon \tilde{\mathbf{Z}}; \mathcal{B})}{2\varepsilon}, \quad (3)$$

where $\tilde{\mathbf{Z}} = \{\tilde{\mathbf{Z}}_i\}_{i=1}^l$ denotes the set of layer-wise perturbations. The gradient estimate for layer i is then $\hat{\nabla}_{\mathbf{W}_i} \mathcal{L} = \rho \tilde{\mathbf{Z}}_i$, and parameters are updated via SGD: $\mathbf{W}_i \leftarrow \mathbf{W}_i - \eta \hat{\nabla}_{\mathbf{W}_i} \mathcal{L}$.

By Lemma 1 of (Yu et al., 2025), the vectorized perturbation $\text{vec}(\tilde{\mathbf{Z}}_i) = (\mathbf{V}_i \otimes \mathbf{U}_i) \text{vec}(\mathbf{Z}_i)$ where $\mathbf{V}_i \otimes \mathbf{U}_i$ is a column-orthonormal matrix of size $m_i n_i \times r^2$. Thus, SubZero effectively performs ZO gradient estimation in an r^2 -dimensional subspace per layer, reducing variance from $O(m_i n_i)$ to $O(r^2)$. The total effective subspace dimension is $q = l r^2 \ll d$.

Limitations of SubZero. Despite its advantages, SubZero has three key limitations that motivate SubZero+. First, it uses only a *single query* ($K = 1$) per iteration, leaving substantial variance reduction on the table. Second, it employs vanilla SGD in the compressed subspace, missing the opportunity for per-layer adaptive optimization. Third, its QR-based subspace construction is implementation-dependent.

4 THE SUBZERO+ FRAMEWORK

SubZero+ extends the SubZero framework along three synergistic axes: multi-query subspace gradient estimation (Sec. 4.1), low-rank subspace Adam (Sec. 4.2), and QR-based subspace construction with sign correction (Sec. 4.3). These components form a dependency chain: the QR-based subspace construction with sign correction improves subspace stability; multi-query estimation provides low-noise gradient norms that better capture per-layer curvature; and subspace Adam then leverages these signals to determine adaptive step sizes. Notably, both multi-query estimation and low-rank subspace Adam operate entirely within an $r \times r$ subspace, so all persistent optimizer state scales as $O(r^2)$ per layer. We describe each component in turn and then present the complete algorithm.

4.1 MULTI-QUERY SUBSPACE GRADIENT ESTIMATION

SubZero estimates the gradient from a single perturbation, leaving its variance at $O(r^2)$ per layer. In principle, averaging over $K > 1$ independent perturbations should reduce this by a factor of K . However, naive multi-query averaging in the full space suffers from the *multi-query paradox* (Lin et al., 2025): the per-iteration variance reduction is offset by the $K \times$ increase in query cost. Our key insight is that the subspace structure of SubZero naturally resolves this paradox: when multiple queries share the same low-rank subspace (defined by fixed projection matrices $\mathbf{U}_i, \mathbf{V}_i$), their gradient estimates are aligned along the subspace basis directions. Averaging across queries reinforces signal components within the subspace while canceling orthogonal noise components, yielding variance reduction that translates into genuine optimization improvement. Critically, subspace-based fusion avoids the $O(d)$ auxiliary state that full-space multi-query averaging would require, preserving the defining advantage of ZO optimization—inference-level memory (see Figure 2).

Formally, let $\mathbf{U}_i \in \mathbb{R}^{m_i \times r}$ and $\mathbf{V}_i \in \mathbb{R}^{n_i \times r}$ be fixed projection matrices for layer i at the current step. For each query $k = 1, \dots, K$, we independently sample $\mathbf{Z}_i^{(k)} \in \mathbb{R}^{r \times r}$ with i.i.d. $\mathcal{N}(0, 1)$ entries, and construct the full-space perturbation:

$$\tilde{\mathbf{Z}}_i^{(k)} = \mathbf{U}_i \mathbf{Z}_i^{(k)} \mathbf{V}_i^\top. \quad (4)$$

Let $\tilde{\mathbf{Z}}^{(k)} = \{\tilde{\mathbf{Z}}_i^{(k)}\}_{i=1}^l$ denote the set of perturbations across all l layers. We compute the baseline loss $\ell_0 = \mathcal{L}(\mathcal{W}; \mathcal{B})$ and the perturbed loss for each query:

$$\ell_k = \mathcal{L}(\mathcal{W} + \varepsilon \tilde{\mathbf{Z}}^{(k)}; \mathcal{B}). \quad (5)$$

Using the forward-difference scheme, the multi-query gradient estimate for layer i is:

$$\hat{\nabla}_{\mathbf{W}_i} \mathcal{L}(\mathcal{W}; \mathcal{B}) = \frac{1}{K} \sum_{k=1}^K \frac{\ell_k - \ell_0}{\varepsilon} \tilde{\mathbf{Z}}_i^{(k)} = \frac{1}{K\varepsilon} \mathbf{U}_i \left(\sum_{k=1}^K (\ell_k - \ell_0) \mathbf{Z}_i^{(k)} \right) \mathbf{V}_i^\top. \quad (6)$$

其中 $\mathbf{Z}_i \in \mathbb{R}^{r \times r}$ 是一个低维随机矩阵，其元素为独立同分布的 $\mathcal{N}(0, 1)$ 。损失差异在全局计算：

$$\rho = \frac{\mathcal{L}(\mathcal{W} + \varepsilon \tilde{\mathbf{Z}}; \mathcal{B}) - \mathcal{L}(\mathcal{W} - \varepsilon \tilde{\mathbf{Z}}; \mathcal{B})}{2\varepsilon}, \quad (3)$$

其中 $\tilde{\mathbf{Z}} = \{\tilde{\mathbf{Z}}_i\}_{i=1}^l$ 表示按层的扰动集合。第 i 层的梯度估计为 $\hat{\nabla}_{\mathbf{W}_i} \mathcal{L} = \rho \tilde{\mathbf{Z}}_i$ ，然后参数通过 SGD 更新： $\mathbf{W}_i \leftarrow \mathbf{W}_i - \eta \hat{\nabla}_{\mathbf{W}_i} \mathcal{L}$ 。

根据 (Yu 等, 2025) 的引理 1，向量化扰动 $\text{vec}(\mathbf{Z}_i) = (\mathbf{V}_i \otimes \mathbf{U}_i) \text{vec}(\mathbf{Z}_i)$ ，其中 $\mathbf{V}_i \otimes \mathbf{U}_i$ 是一个大小为 $m_i n_i \times r^2$ 的列正交矩阵。因此，SubZero 实际上在每一层执行 r^2 维子空间的零阶梯度估计，将方差从 $O(m_i n_i)$ 降低到 $O(r^2)$ 。总的有效子空间维度为 $q = l r^2 \ll d$ 。

SubZero 的局限性。尽管具有优势，SubZero 仍存在三个主要局限，这促使了 SubZero+ 的发展。首先，它在每次迭代中仅使用单个查询 ($K = 1$)，从而在方差减少方面留有很大的提升空间。其次，它在压缩子空间中使用普通的 SGD，错过了对每一层进行自适应优化的机会。第三，它基于 QR 的子空间构建依赖于具体实现。

4 THE SUBZERO+ 框架

SubZero+ 沿着三个协同轴扩展了 SubZero 框架：多查询子空间梯度估计（第 4.1 节）、低秩子空间 Adam（第 4.2 节）以及具有符号校正的基于 QR 的子空间构建（第 4.3 节）。这些组件形成了一个依赖链：具有符号校正的基于 QR 的子空间构建提高了子空间的稳定性；多查询估计提供了低噪声梯度范数，更好地捕捉每一层的曲率；然后子空间 Adam 利用这些信号来确定自适应步长。值得注意的是，多查询估计和低秩子空间 Adam 都完全在 $r \times r$ 子空间中操作，因此所有持久的优化器状态在每层的规模为 $O(r^2)$ 。我们将依次描述每个组件，然后介绍完整算法。

4.1 多查询子空间梯度估计

SubZero 从单次扰动中估计梯度，使每层的方差保持在 $O(r^2)$ 。原则上，对 $K > 1$ 次独立扰动进行平均应将其降低 K 倍。然而，在整个空间中进行天真的多查询平均会受到多查询悖论 (Lin 等, 2025) 的影响：每次迭代的方差降低被查询成本的 $K \times$ 增加所抵消。我们的关键见解是 SubZero 的子空间结构自然地解决了这一悖论：当多个查询共享相同的低秩子空间（由固定投影矩阵 $\mathbf{U}_i, \mathbf{V}_i$ 定义）时，它们的梯度估计沿子空间基方向对齐。在查询间进行平均会强化子空间内的信号成分，同时抵消正交的噪声成分，从而实现方差降低，并转化为实际的优化改进。关键是，基于子空间的融合避免了完整空间多查询平均所需的 $O(d)$ 辅助状态，prese

正式地，设 $\mathbf{U}_i \in \mathbb{R}^{m_i \times r}$ 和 $\mathbf{V}_i \in \mathbb{R}^{n_i \times r}$ 是当前步骤中第 i 层的固定投影矩阵。对于每个查询 $k = 1, \dots, K$ ，我们独立采样 $\mathbf{Z}_i^{(k)} \in \mathbb{R}^{r \times r}$ ，其条目为独立同分布的 $\mathcal{N}(0, 1)$ ，并构造全空间扰动：

$$\tilde{\mathbf{Z}}_i^{(k)} = \mathbf{U}_i \mathbf{Z}_i^{(k)} \mathbf{V}_i^\top. \quad (4)$$

令 $\tilde{\mathbf{Z}}^{(k)} = \{\tilde{\mathbf{Z}}_i^{(k)}\}_{i=1}^l$ 表示所有 l 层的扰动集合。我们计算基线损失 $\ell_0 = \mathcal{L}(\mathcal{W}; \mathcal{B})$ 以及每个查询的扰动损失：

$$\ell_k = \mathcal{L}(\mathcal{W} + \varepsilon \tilde{\mathbf{Z}}^{(k)}; \mathcal{B}). \quad (5)$$

使用前向差分公式，层 i 的多查询梯度估计为：

$$\hat{\nabla}_{\mathbf{W}_i} \mathcal{L}(\mathcal{W}; \mathcal{B}) = \frac{1}{K} \sum_{k=1}^K \frac{\ell_k - \ell_0}{\varepsilon} \tilde{\mathbf{Z}}_i^{(k)} = \frac{1}{K\varepsilon} \mathbf{U}_i \left(\sum_{k=1}^K (\ell_k - \ell_0) \mathbf{Z}_i^{(k)} \right) \mathbf{V}_i^\top. \quad (6)$$

From variance reduction to curvature awareness. Beyond variance reduction, multi-query estimation unlocks a signal that single-query estimation is too noisy to provide: the per-layer gradient norm $\|\bar{\mathbf{Z}}_i^t\|_F$, where $\bar{\mathbf{Z}}_i^t = \frac{1}{K} \sum_{k=1}^K \rho_k^t \mathbf{Z}_i^{(k)}$ is the aggregated low-rank gradient. With multiple queries, this norm becomes a reliable proxy for local loss landscape curvature, serving as the foundation for the per-layer adaptive optimization developed next (Figure 3 provides a 2D illustration).

4.2 LOW-RANK SUBSPACE ADAM

SubZero applies vanilla SGD in the compressed subspace with a uniform learning rate η across all layers. However, the per-layer multi-query gradient norm $\gamma_i^t = \|\bar{\mathbf{Z}}_i^t\|_F$ can differ by an order of magnitude across layers. A uniform learning rate treats a layer in a flat basin identically to one on a sharp ridge, resulting in either overly conservative updates (slowing convergence) or dangerously aggressive ones (risking divergence).

We address this by running the Adam optimizer (Kingma & Ba, 2015) entirely within the $r \times r$ low-rank space. For each layer i , we maintain first- and second-moment estimates $\mathbf{M}_i^t, \mathbf{V}_i^t \in \mathbb{R}^{r \times r}$:

$$\mathbf{M}_i^t = \beta_1 \mathbf{M}_i^{t-1} + (1 - \beta_1) \bar{\mathbf{Z}}_i^t, \quad (7)$$

$$\mathbf{V}_i^t = \beta_2 \mathbf{V}_i^{t-1} + (1 - \beta_2) (\bar{\mathbf{Z}}_i^t \odot \bar{\mathbf{Z}}_i^t), \quad (8)$$

where $\odot$ denotes element-wise multiplication and $\beta_1, \beta_2 \in [0, 1]$ are decay rates. After bias correction

$$\hat{\mathbf{M}}_i^t = \mathbf{M}_i^t / (1 - \beta_1^t), \quad \hat{\mathbf{V}}_i^t = \mathbf{V}_i^t / (1 - \beta_2^t), \quad (9)$$

the parameter update is:

$$\mathbf{W}_i^{t+1} = \mathbf{W}_i^t - \eta \mathbf{U}_i^t \left(\frac{\hat{\mathbf{M}}_i^t}{\sqrt{\hat{\mathbf{V}}_i^t} + \epsilon_{\text{adam}}} \right) (\mathbf{V}_i^t)^\top, \quad (10)$$

where division and square-root are element-wise. The element-wise preconditioning by $1/\sqrt{\hat{\mathbf{V}}_i^t}$ automatically assigns smaller effective step sizes to layers with large gradient magnitudes (steep curvature) and larger steps to layers with small magnitudes (flat regions). Both moment buffers reside in $\mathbb{R}^{r \times r}$, contributing only $2r^2$ floats per layer, negligible relative to the model size.

Figure 2 quantifies this advantage for FT of OPT-13B on SST-2. MeZO’s memory grows from 25.9 GB (single-query SGD) to 97.2 GB (multi-query Adam)—a $3.8\times$ increase—due to $O(d)$ gradient accumulation and moment buffers. SubZero+ runs the same multi-query Adam configuration at only 27.9 GB, because all auxiliary states are confined to $r \times r$ subspace buffers. All four SubZero+ variants (SGD or Adam, single- or multi-query) consume identical memory, underscoring that the subspace design decouples optimizer sophistication from memory cost.

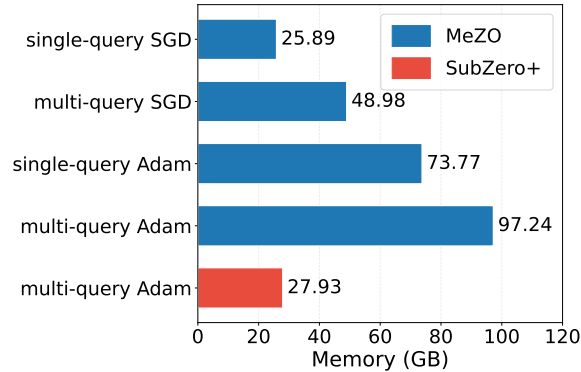


Figure 2: GPU memory breakdown on OPT-13B.

4.3 IMPLEMENTATION AND ALGORITHM

Why Multi-Query and Adam Must Work Together. Figure 3 illustrates this synergy on a 2D loss landscape. With $K = 1$ (Figure 3(b)), the gradient norm field is dominated by noise. With $K = 100$ (Figure 3(c)), it faithfully reflects local curvature. Subspace Adam exploits this signal to assign adaptive step sizes—small on steep walls, large on flat regions—enabling near-monotone descent.

QR Subspace Construction with Sign Correction. The projection matrices $\mathbf{U}_i, \mathbf{V}_i$ are regenerated every F steps by drawing random Gaussian matrices $\Omega_U \in \mathbb{R}^{m_i \times r}, \Omega_V \in \mathbb{R}^{n_i \times r}$ and computing

预印本。

从方差减少到曲率感知。超越方差减少，多查询估计解锁了单次查询估计因噪声过大而无法提供的信号：每层梯度范数 $\|\mathbf{Z}_i^t\|_F$ ，其中 $\mathbf{Z}_i^t = \frac{1}{K} \sum_{k=1}^K \rho_k^t \mathbf{Z}_i^{(k)}$ 是聚合的低秩梯度。通过多次查询，这个范数成为局部损失曲面曲率的可靠代理，为接下来开发的每层自适应优化奠定了基础（图3提供了二维示意图）。

4.2 低秩子空间ADAM

SubZero 在压缩子空间中应用标准 SGD，并在所有层使用统一学习率 η 。然而，每层的多查询梯度范数 $\gamma_i^t = \|\mathbf{Z}_i^t\|_F$ 在各层之间可能相差一个数量级。统一学习率会将位于平坦盆地的层与位于陡峭脊上的层同等对待，导致更新过于保守（降低收敛速度）或过于激进（存在发散风险）。

我们通过完全在 $r \times r$ 低秩空间中运行 Adam 优化器 (Kingma & Ba, 2015) 来解决这个问题。对于每一层 i ，我们保持一阶和二阶矩的估计 $\mathbf{M}_i^t, \mathbf{V}_i^t \in \mathbb{R}^{r \times r}$:

$$\mathbf{M}_i^t = \beta_1 \mathbf{M}_i^{t-1} + (1 - \beta_1) \bar{\mathbf{Z}}_i^t, \quad (7)$$

$$\mathbf{V}_i^t = \beta_2 \mathbf{V}_i^{t-1} + (1 - \beta_2) (\bar{\mathbf{Z}}_i^t \odot \bar{\mathbf{Z}}_i^t), \quad (8)$$

其中 $\odot$ 表示逐元素相乘， $\beta_1, \beta_2 \in [0, 1]$ 是衰减率。进行偏置校正后

$$\hat{\mathbf{M}}_i^t = \mathbf{M}_i^t / (1 - \beta_1^t), \quad \hat{\mathbf{V}}_i^t = \mathbf{V}_i^t / (1 - \beta_2^t), \quad (9)$$

参数更新为:

$$\mathbf{W}_i^{t+1} = \mathbf{W}_i^t - \eta \mathbf{U}_i^t \left(\frac{\hat{\mathbf{M}}_i^t}{\sqrt{\hat{\mathbf{V}}_i^t} + \epsilon_{\text{adam}}} \right) (\mathbf{V}_i^t)^\top, \quad (10)$$

除法和平方根是按元素进行的。通过 $1/\sqrt{\hat{\mathbf{V}}_i^t}$ 进行的按元素预处理会自动将较小的有效步长分配给梯度幅值较大的层（曲率陡峭），并将较大的步长分配给幅值较小的层（平坦区域）。两个动量缓冲区都存在于 $\mathbb{R}^{r \times r}$ 中，每层仅占用 $2r^2$ 个浮点数，相对于模型大小可以忽略不计。

图2量化了OPT-13B在SST-2上的FT的这一优势。MeZO的内存从25.9 GB（单查询SGD）增长到97.2 GB（多查询Adam）——3.8×的增加——这是由于 $O(d)$ 梯度累积和动量缓冲造成的。SubZero+ 在相同的多查询Adam配置下仅使用27.9 GB，因为所有辅助状态都被限制在{subspace}缓冲区。所有四个SubZero+ 变体（SGD或Adam，单查询或多查询）使用相同的内存，这强调了子空间设计将优化器复杂性与内存成本解耦。

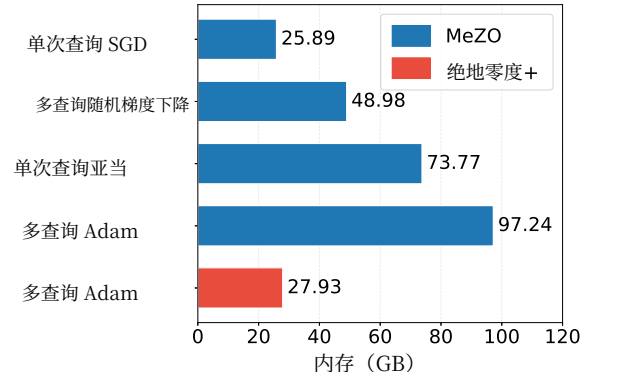


图 2: OPT-13B 的 GPU 内存分解。

4.3 实现与算法

为什么多查询和 Adam 必须一起工作。图 3 展示了这种在二维损失景观上的协同作用。使用 $K = 1$ （图 3(b)），梯度范数场被噪声主导。使用 $K = 100$ （图 3(c)），它能够真实地反映局部曲率。子空间 Adam 利用这一信号分配自适应步长——陡峭区域小，平坦区域大——从而实现近单调下降。

带符号校正的 QR 子空间构建。投影矩阵 $\mathbf{U}_i, \mathbf{V}_i$ 每 F 步通过生成随机高斯矩阵 $\Omega_U \in \mathbb{R}^{m_i \times r}, \Omega_V \in \mathbb{R}^{n_i \times r}$ 并计算来重新生成

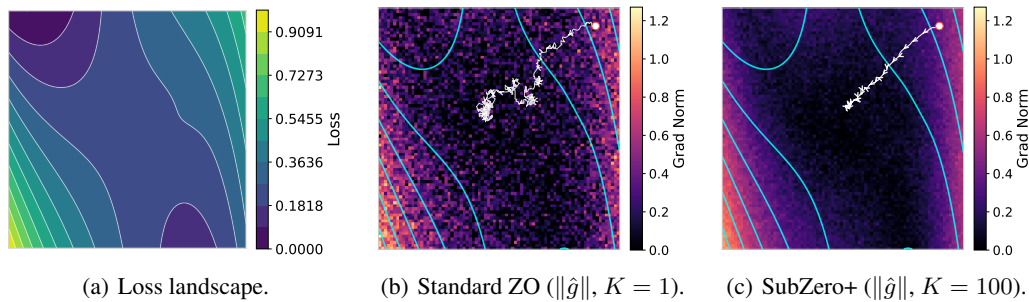


Figure 3: Multi-query and Adam are synergistic. (a) A 2D loss surface with steep walls and a flat valley. (b) Standard ZO gradient norm ($K = 1$): noise dominates, obscuring curvature. (c) SubZero+ gradient norm ($K = 100$): curvature structure emerges, enabling adaptive step sizes. All panels share the same domain and starting point.

their QR decompositions. Standard QR implementations (e.g., `torch.linalg.qr`) do not guarantee a positive diagonal on the triangular factor R , leaving the column signs of Q arbitrary among 2^r choices. Only the unique sign configuration satisfying $R_{ii} > 0$ produces a Haar-distributed Q , which is rotationally invariant and directionally unbiased (Mezzadri, 2007). We enforce this via a simple post-hoc correction:

$$U_i = \tilde{U}_i \text{diag}(\text{sgn}(\tilde{R}_{U,11}), \dots, \text{sgn}(\tilde{R}_{U,rr})), V_i = \tilde{V}_i \text{diag}(\text{sgn}(\tilde{R}_{V,11}), \dots, \text{sgn}(\tilde{R}_{V,rr})), \quad (11)$$

which multiplies each column of $\tilde{Q}$ by the sign of the corresponding $\tilde{R}$ diagonal entry. This correction restores the missing sign symmetry and makes the basis distribution invariant to implementation-dependent QR sign conventions. We apply this procedure independently to the left and right projection matrices. The cost is $O(r)$ column sign flips per layer, negligible compared to the $O(m_i n_i r)$ QR cost. See Appendix A.3 for empirical validation.

Complete Algorithm. Algorithm 1 constructs the Haar-corrected projection matrices, Algorithm 2 generates and applies perturbations, and Algorithm 3 gives the full SubZero+ training loop. Each iteration requires $K + 1$ forward passes—one baseline and K perturbed—and remains entirely BP-free. SubZero+ inherits SubZero’s compatibility with full-parameter tuning (FT), LoRA (Hu et al., 2022), prefix tuning (Li & Liang, 2021), and prompt tuning (Lester et al., 2021). The multi-query estimation and low-rank Adam are applied to the trainable parameters in each scheme.

5 EXPERIMENTS

We conduct comprehensive experiments to evaluate SubZero+ against state-of-the-art ZO baselines across diverse models, tasks, and fine-tuning schemes. Due to space constraints, detailed hyperparameter configurations and additional results are provided in Appendix A.2.

5.1 EXPERIMENTAL SETUP

Comprehensive Validation. We evaluate autoregressive (decoder-only) LLMs including OPT-1.3B, OPT-13B (Zhang et al., 2022), OPT-30B (Zhang et al., 2022), LLaMA3.1-8B (Team & AI@Meta, 2024), and Qwen2.5-32B (Team, 2024). Tasks span the SuperGLUE benchmark (Wang et al., 2019), covering classification, multiple-choice, and generation tasks. We cover the FT and LoRA ($r = 8$, $\alpha = 16$) schemes. We compare against: MeZO (Malladi et al., 2023) and SubZero (Yu et al., 2025). For FO reference, we include Adam (FT). Zero-shot inference serves as a lower bound.

Hyperparameters. Default settings for SubZero+: queries $K = 99$, Adam parameters $\beta_1 = 0.9$, $\beta_2 = 0.95$, and $\epsilon_{\text{adam}} = 10^{-8}$. Subspace rank r and update frequency F are specified in Appendix A.2. Learning rates and perturbation scales are tuned via grid search per benchmark (see Appendix A.2). All ZO methods use batch size 16 and select the best checkpoint by validation loss, following (Malladi et al., 2023). MeZO and SubZero employ two-point gradient estimation, consuming 2 forward passes per step, for a total forward-pass budget of 40K over 20K steps. SubZero+ uses $K + 1$ forward passes per step (K perturbed and one baseline). For fair comparison, we adjust

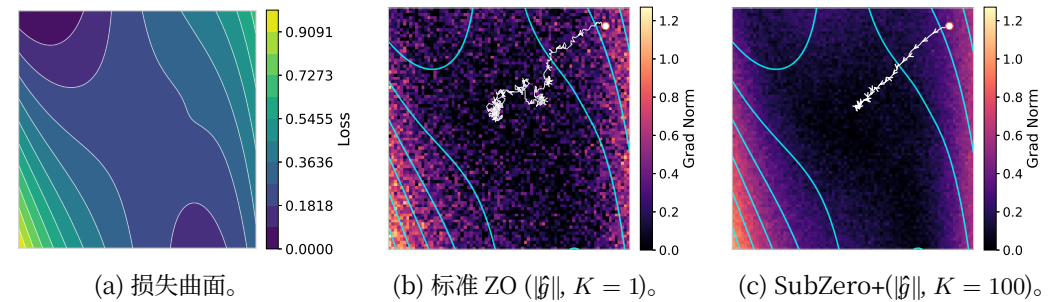


图 3: 多查询和 Adam 是协同作用的。(a) 具有陡峭墙壁和平坦山谷的二维损失表面。(b) 标准 ZO 梯度范数 ($K = 1$): 噪声占主导, 掩盖了曲率。(c) SubZero+ 梯度范数 ($K = 100$): 曲率结构显现, 支持自适应步长。所有面板共享相同的域和起点。

它们的 QR 分解。标准的 QR 实现 (例如 `torch.linalg.qr`) 不能保证三角因子 R 的对角线为正, 使得 Q 的列符号在 2^r 个选择中任意。只有满足 $R_{ii} > 0$ 的唯一符号配置才能生成 Haar 分布的 Q , 这种分布具有旋转不变性且方向上无偏 (Mezzadri, 2007)。我们通过一个简单的事后修正来实现这一点:

$$U_i = \tilde{U}_i \text{diag}(\text{sgn}(\tilde{R}_{U,11}), \dots, \text{sgn}(\tilde{R}_{U,rr})), V_i = \tilde{V}_i \text{diag}(\text{sgn}(\tilde{R}_{V,11}), \dots, \text{sgn}(\tilde{R}_{V,rr})), \quad (11)$$

将 Q 的每一列乘以对应 R 对角元素的符号。此修正恢复了缺失的符号对称性, 并使基分布对实现相关的 QR 符号约定保持不变。我们将此过程独立应用于左投影矩阵和右投影矩阵。成本为每层 $O(r)$ 列符号翻转, 相较于 $O(m_i n_i r)$ 的 QR 成本可以忽略不计。经验验证见附录 A.3。

完整算法。算法1构建Haar校正投影矩阵, 算法2生成并应用扰动, 算法3给出完整的SubZero+训练循环。每次迭代需要 $K + 1$ 次前向传递——一次基线和 K 次扰动——并且完全无需反向传播。SubZero+ 继承了SubZero在全参数调优 (FT)、LoRA (Hu等, 2022)、前缀调优 (Li & Liang, 2021) 和提示调优 (Lester等, 2021) 方面的兼容性。多查询估计和低秩Adam被应用于每种方案中的可训练参数。

5 个实验

我们进行了全面的实验, 以评估 SubZero+ 在各种模型、任务和微调方案上相对于最先进的零阶基线的表现。由于篇幅限制, 详细的超参数配置和额外结果见附录 A.2。

5.1 实验设置

全面验证。我们评估自回归 (仅解码器) 大语言模型 (LLM), 包括 OPT-1.3B、OPT-13B (Zhang 等, 2022)、OPT-30B (Zhang 等, 2022)、LLaMA3.1-8B (Team & AI@Meta, 2024) 和 Qwen2.5-32B (Team, 2024)。任务涵盖 SuperGLUE 基准 (Wang 等, 2019), 包括分类、多项选择和生成任务。我们涵盖 FT 和 LoRA ($r = 8, \alpha = 16$) 方案。我们进行对比的模型包括: MeZO (Malladi 等, 2023) 和 SubZero (Yu 等, 2025)。作为参考, 我们包括 Adam (FT)。零样本推理作为下界。

超参数。SubZero+的默认设置: 查询数 $K = 99$, Adam参数 $\beta_1 = 0.9$, $\beta_2 = 0.95$ 和 $\epsilon_{\text{adam}} = 10^{-8}$ 。子空间秩 r 和更新频率 F 在附录 A.2 中说明。学习率和扰动尺度通过每个基准的网格搜索进行调整 (见附录 A.2)。所有 ZO 方法使用批量大小 16, 并通过验证损失选择最佳检查点, 遵循 (Malladi 等, 2023)。MeZO 和 SubZero 采用两点梯度估计, 每步消耗 2 次前向传播, 总前向传播预算为 20K 步中的 40K 次步数。SubZero+每步使用 $K + 1$ 次前向传播 (K 次扰动和一次基线)。为了公平比较, 我们调整

Algorithm 1 HaarProjMatrix(r, m_i, n_i)	Algorithm 2 PerturbParams($\mathcal{W}, \mathcal{U}, \mathcal{V}, r, \varepsilon, s$)
Input: Rank r , dimensions m_i, n_i . 1: Draw $\Omega_U \in \mathbb{R}^{m_i \times r}, \Omega_V \in \mathbb{R}^{n_i \times r}$ with i.i.d. $\mathcal{N}(0, 1)$ 2: $\tilde{U}_i, R_U \leftarrow \text{QR}(\Omega_U); \tilde{V}_i, R_V \leftarrow \text{QR}(\Omega_V)$ 3: $s_U \leftarrow [\text{sgn}(R_{U,11}), \dots, \text{sgn}(R_{U,rr})]^\top$; $s_V \leftarrow [\text{sgn}(R_{V,11}), \dots, \text{sgn}(R_{V,rr})]^\top$ 4: $U_i \leftarrow \tilde{U}_i \text{diag}(s_U); V_i \leftarrow \tilde{V}_i \text{diag}(s_V)$ 5: return U_i, V_i	Input: Model parameters $\mathcal{W}$, projection sets $\mathcal{U}, \mathcal{V}$, rank r , perturbation scale ε , seed s . 1: Reset RNG with seed s 2: $\mathcal{Z} \leftarrow \emptyset$ 3: for $i = 1, 2, \dots, l$ do 4: Generate $Z_i \in \mathbb{R}^{r \times r}$ with i.i.d. $\mathcal{N}(0, 1)$ entries 5: $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{Z_i\}$ 6: $W_i \leftarrow W_i + \varepsilon U_i Z_i V_i^\top$ 7: return $\mathcal{W}, \mathcal{Z}$
Algorithm 3 SubZero+	
Input: Parameter matrices $W_i \in \mathbb{R}^{m_i \times n_i}$ for $i = 1, \dots, l$, loss $\mathcal{L}$, step budget T , perturbation scale ε , learning rate η , subspace change frequency F , rank r , query count K , Adam parameters $\beta_1, \beta_2, \epsilon_{\text{adam}}$. 1: Initialize $M_i^0 \leftarrow \mathbf{0}, V_i^0 \leftarrow \mathbf{0}$ for all i $\{r \times r$ buffers} 2: for $t = 0, 1, \dots, T - 1$ do 3: Sample a minibatch $\mathcal{B}^t \subset \mathcal{D}$ 4: for $i = 1, 2, \dots, l$ do 5: if $t \bmod F \equiv 0$ then 6: $U_i^t, V_i^t \leftarrow \text{HaarProjMatrix}(r, m_i, n_i)$ 7: else 8: $U_i^t \leftarrow U_i^{t-1}, V_i^t \leftarrow V_i^{t-1}$ 9: Compute baseline loss: $\ell_0^t \leftarrow \mathcal{L}(\mathcal{W}^t; \mathcal{B}^t)$ 10: Initialize $\bar{Z}_i^t \leftarrow \mathbf{0} \in \mathbb{R}^{r \times r}$ for all i 11: for $k = 1, 2, \dots, K$ do 12: Sample a random seed s_k^t 13: $\mathcal{W}^t, \mathcal{Z}_k^t \leftarrow \text{PerturbParams}(\mathcal{W}^t, \mathcal{U}^t, \mathcal{V}^t, r, \varepsilon, s_k^t)$ 14: $\ell_k^t \leftarrow \mathcal{L}(\mathcal{W}^t; \mathcal{B}^t)$ 15: $\mathcal{W}^t, _ \leftarrow \text{PerturbParams}(\mathcal{W}^t, \mathcal{U}^t, \mathcal{V}^t, r, -\varepsilon, s_k^t)$ 16: $\rho_k^t \leftarrow (\ell_k^t - \ell_0^t) / \varepsilon$ 17: for $i = 1, 2, \dots, l$ do 18: $\bar{Z}_i^t \leftarrow \bar{Z}_i^t + \rho_k^t \cdot Z_{k,i}^t$ 19: for $i = 1, 2, \dots, l$ do 20: $\bar{Z}_i^t \leftarrow \bar{Z}_i^t / K$ 21: Update W_i^{t+1} via Adam (Eqs. (7)–(10)) with gradient $\bar{Z}_i^t$ 22: return $\mathcal{W}^T$	

SubZero+’s step count so that its total forward-pass budget matches the 40K budget; with the default $K = 99$, SubZero+ runs for 400 steps, consuming $99 + 1 = 100$ forwards per step, matching the $2 \times 20K = 40K$ total budget of MeZO and SubZero.

5.2 MAIN RESULTS

5.2.1 AUTO-REGRESSIVE LLMs UNDER FT SCHEME

We first evaluate on three autoregressive models of varying scales: OPT-1.3B, LLaMA3.1-8B, and OPT-13B. Following prior ZO work (Malladi et al., 2023; Zhao et al., 2025), we use the FT scheme for all ZO methods. Results are presented in Table 1. SubZero+ consistently outperforms all ZO baselines across all three model scales. On OPT-1.3B, SubZero+ achieves 66.3% average accuracy across 9 tasks, outperforming SubZero (65.5%) and MeZO (64.6%). On LLaMA3.1-8B, SubZero+ reaches 83.3%, surpassing SubZero (82.5%) and MeZO (82.3%). On OPT-13B, SubZero+ attains 73.0% average accuracy, again improving over SubZero (70.3%) and MeZO (69.9%). On LLaMA3.1-8B, subspace SGD marginally outperforms subspace Adam (83.7% vs. 83.3%), attributable to different learning rate search spaces. Representative loss trajectories are illustrated in Figure 4.

算法 1 HaarProjMatrix(r, m_i, n_i)	算法 2 扰动参数($\mathcal{W}, \mathcal{U}, \mathcal{V}, r, \varepsilon, s$)
输入: 秩 r , 维度 m_i, n_i 1: 绘制 $\Omega_U \in \mathbb{R}^{m_i \times r}, \Omega_V \in \mathbb{R}^{n_i \times r}$ 与独立同分布 $\mathcal{N}(0, 1)$ 2: $\tilde{U}_i, R_U \leftarrow \text{QR}(\Omega_U); \tilde{V}_i, R_V \leftarrow \text{QR}(\Omega_V)$ 3: $s_U \leftarrow [\text{sgn}(R_{U,11}) \dots, \text{sgn}(R_{U,rr})]^\top$; $s_V \leftarrow [\text{sgn}(R_{V,11}) \dots, \text{sgn}(R_{V,rr})]^\top$ 4: $U_i \leftarrow \tilde{U}_i \text{diag}(s_U); V_i \leftarrow \tilde{V}_i \text{diag}(s_V)$ 5: 返回 U_i, V_i	输入: 模型参数 $\mathcal{W}$, 投影集 $\mathcal{U}, \mathcal{V}$, 等级 r , 扰动规模 ε , 种子 s . 1: 使用种子 s 重置随机数生成器 2: $\mathcal{Z} \leftarrow \emptyset$ 3: 对于 $i = 1, 2, \dots, l$ 做 4: 生成 $Z_i \in \mathbb{R}^{r \times r}$, 其条目为独立同分布的 $\mathcal{N}(0, 1)$ 5: $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{Z_i\}$ 6: $W_i \leftarrow W_i + \varepsilon U_i Z_i V_i^\top$ 7: 返回 $\mathcal{W}, \mathcal{Z}$
算法 3 SubZero+	
输入: 参数矩阵 $W_i \in \mathbb{R}^{m_i \times n_i}$ 对应 $i = 1, \dots, l$, 损失 $\mathcal{L}$, 步长预算 T , 干扰尺度 ε , 学习率 η , 子空间更改频率 F , 等级 r , 查询次数 K , Adam 参数 $\beta_1, \beta_2, \epsilon_{\text{adam}}$. 1: 初始化 $M_i^0 \leftarrow \mathbf{0}, V_i^0 \leftarrow \mathbf{0}$ 对所有 i $\{r \times r$ 缓冲区} 2: 对于 $t = 0, 1, \dots, T - 1$ 做 3: 抽取一个小批量 $\mathcal{B}^t \subset \mathcal{D}$ 4: 为 $i = 1, 2, \dots, l$ 做 5: 如果 $t \bmod F \equiv 0$ 那么 6: $U_i^t, V_i^t \leftarrow \text{Haar投影矩阵}(r, m_i, n_i)$ 7: else 8: $U_i^t \leftarrow U_i^{t-1}, V_i^t \leftarrow V_i^{t-1}$ 9: 计算基线损失: $\ell_0^t \leftarrow \mathcal{L}(\mathcal{W}^t; \mathcal{B}^t)$ 10: 初始化 $\bar{Z}_i^t \leftarrow \mathbf{0} \in \mathbb{R}^{r \times r}$ 适用于所有 i 11: 对于 $k = 1, 2, \dots, K$ 做 12: 抽取一个随机种子 s_k^t 13: $\mathcal{W}^t, \mathcal{Z}_k^t \leftarrow \text{扰动参数}(\mathcal{W}^t, \mathcal{U}^t, \mathcal{V}^t, r, \varepsilon, s_k^t)$ 14: $\ell_k^t \leftarrow \mathcal{L}(\mathcal{W}^t; \mathcal{B}^t)$ 15: $\mathcal{W}^t, _ \leftarrow \text{扰动参数}(\mathcal{W}^t, \mathcal{U}^t, \mathcal{V}^t, r, -\varepsilon, s_k^t)$ 16: $\rho_k^t \leftarrow (\ell_k^t - \ell_0^t) / \varepsilon$ 17: 为 $i = 1, 2, \dots, l$ 做 18: $\bar{Z}_i^t \leftarrow \bar{Z}_i^t + \rho_k^t \cdot Z_{k,i}^t$ 19: 为 $i = 1, 2, \dots, l$ 做 20: $\bar{Z}_i^t \leftarrow \bar{Z}_i^t / K$ 21: 使用梯度 $\bar{Z}_i^t$ 通过 Adam (方程 (7)–(10)) 更新 W_i^{t+1} 22: 返回 $\mathcal{W}^T$	

调整SubZero+的步数, 使其总前向传递预算与40K预算相匹配; 使用默认 $K = 99$ 时, SubZero+ 运行400步, 每步消耗 $99 + 1 = 100$ 个前向, 与MeZO和SubZero的 $2 \times 20K = 40K$ 总预算相匹配。

5.2 主要结果

5.2.1 在 FT 方案下的自回归大型语言模型

E

我们首先在三种不同规模的自回归模型上进行评估: OPT-1.3B、LLaMA3.1-8B 和 OPT-13B。遵循之前的 ZO 工作 (Malladi 等, 2023; Zhao 等, 2025), 我们对所有 ZO 方法使用 FT 方案。结果如表 1 所示。SubZero+ 在所有三种模型规模上持续优于所有 ZO 基线。在 OPT-1.3B 上, SubZero+ 在 9 个任务中取得了 66.3% 的平均准确率, 超过了 SubZero (65.5%) 和 MeZO (64.6%)。在 LLaMA3.1-8B 上, SubZero+ 达到 83.3%, 超过 SubZero (82.5%) 和 MeZO (82.3%)。在 OPT-13B 上, SubZero+ 实现了 73.0% 的平均准确率, 再次优于 SubZero (70.3%) 和 MeZO (69.9%)。在 LLaMA3.1-8B 上, 子空间 SGD 略微优于子空间 Adam (83.7% 对 83.3%), 这归因于不同的学习率搜索空间。典型的损失轨迹如图 4 所示。

Table 1: Main results on autoregressive LLMs under the FT scheme. SubZero+ uses $K = 99$ queries. All methods share a 40K forward-pass budget.

Model	Method Task type	SST-2	RTE	BoolQ	WIC	MultiRC	COPA	ReCoRD	SQuAD	DROP	Avg.
		—— Classification ——				— Multi-choice —		— Generation —			
OPT-1.3B	Zero-shot	53.5	53.4	45.5	57.5	45.4	75.0	70.5	27.2	11.1	48.8
	FO (Adam)	92.0	78.0	75.8	66.0	70.9	77.0	71.1	81.7	31.1	71.5
	MeZO	90.2	64.3	65.0	58.3	59.7	74.0	71.0	76.4	22.5	64.6
	SubZero	90.9	67.5	65.4	58.1	59.5	75.0	70.9	76.4	25.4	65.5
	SubZero+/SGD	90.0	60.3	66.4	60.0	61.7	75.0	71.9	77.2	25.4	65.3
LLaMA3.1-8B	Zero-shot	59.8	46.6	76.3	59.6	62.3	64.9	83.7	85.0	28.5	63.0
	FO (Adam)	95.9	88.1	87.7	76.3	89.3	84.0	75.8	91.4	70.1	84.3
	MeZO	91.6	85.9	83.3	66.3	86.7	89.0	82.6	90.3	65.1	82.3
	SubZero	91.2	83.8	83.3	65.5	87.2	90.0	82.6	90.9	67.8	82.5
	SubZero+/SGD	95.4	84.8	84.2	69.6	87.2	92.0	82.0	90.7	67.5	83.7
OPT-13B	Zero-shot	58.8	59.6	59.0	55.0	46.9	80.0	81.2	46.2	14.6	55.7
	FO (Adam)	95.1	83.4	80.6	68.0	78.9	88.0	81.8	87.0	34.7	77.5
	MeZO	91.3	67.5	67.1	59.1	61.5	87.0	81.4	83.6	30.6	69.9
	SubZero	90.8	66.4	68.3	58.3	66.8	88.0	81.2	83.6	29.2	70.3
	SubZero+/SGD	92.8	70.8	74.4	60.7	69.5	88.0	81.2	85.3	30.6	72.5
OPT-13B	SubZero+/Adam	93.5	74.0	75.2	63.2	64.7	89.0	81.7	85.2	30.6	73.0

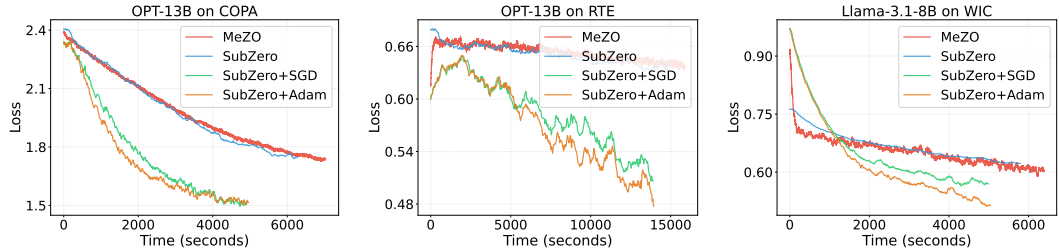


Figure 4: Visualization of selected loss curves in Table 1.

5.2.2 AUTO-REGRESSIVE LLMs UNDER LoRA SCHEME

We further evaluate SubZero+ under the LoRA scheme ($r = 8, \alpha = 16$) on OPT-13B, LLaMA3.1-8B, and OPT-30B. LoRA learns low-rank adaptation matrices for attention projections, and is widely used in practice. We compare the best results from each ZO method across hyperparameter searches. Results are presented in Table 2. As shown in Table 2, SubZero+ outperforms both MeZO and SubZero under LoRA scheme on both model scales. On OPT-13B, SubZero+ achieves 70.7% average accuracy, outperforming MeZO (68.7%) by +2.0% and SubZero (68.6%) by +2.1%. On LLaMA3.1-8B, SubZero+ achieves 83.4% average accuracy, outperforming MeZO (81.2%) by +2.2% and SubZero (81.5%) by +1.9%. For the larger OPT-30B, SubZero+ further boosts the average accuracy to 72.2%, surpassing MeZO (70.1%) by +2.1% and SubZero (69.4%) by +2.8%. The results are consistent with those obtained under the FT scheme.

5.2.3 SCALING TO QWEN2.5-32B

To validate SubZero+ on a substantially larger model, we additionally evaluate on Qwen2.5-32B (Team, 2024) under the FT scheme. We compare against the zero-shot baseline, MeZO, MeZO with LoRA, and SubZero. Results are presented in Table 3.

On Qwen2.5-32B, SubZero+ achieves the highest score on every task, with an average accuracy of 87.4%—outperforming SubZero (85.1%) by +2.3%, MeZO (83.6%) by +3.8%, MeZO-LoRA (76.7%) by +10.7%, and the zero-shot baseline (76.0%) by +11.4%. The gains are particularly large on SST-2 (+5.9% over SubZero) and WIC (+4.7% over SubZero).

表1: 在FT方案下自回归大型语言模型的主要结果。SubZero+ 使用 $K = 99$ 次查询。所有方法使用相同的4万次前向传递预算。

模型	方法 任务类型	SST-2	RTE	BoolQ	WIC	MultiRC	COPA	ReCoRD	SQuAD	DROP	平均
		—— 分类 ——				— 多选 —		— 生成 —			
OPT-1.3B	零样本 FO (亚当)	53.5	53.4	45.5	57.5	45.4	75.0	70.5	27.2	11.1	48.8
		92.0	78.0	75.8	66.0	70.9	77.0	71.1	81.7	31.1	71.5
	MeZO	90.2	64.3	65.0	58.3	59.7	74.0	71.0	76.4	22.5	64.6
	绝地零	90.9	67.5	65.4	58.1	59.5	75.0	70.9	76.4	25.4	65.5
	SubZero+/SGD	90.0	60.3	66.4	60.0	61.7	75.0	71.9	77.2	25.4	65.3
LLaMA3.1-8B	SubZero+/亚当	92.1	65.7	66.9	59.4	61.4	76.0	72.3	77.2	25.4	66.3
	零样本 FO (亚当)	59.8	46.6	76.3	59.6	62.3	64.9	83.7	85.0	28.5	63.0
		95.9	88.1	87.7	76.3	89.3	84.0	75.8	91.4	70.1	84.3
	MeZO	91.6	85.9	83.3	66.3	86.7	89.0	82.6	90.3	65.1	82.3
	绝地零	91.2	83.8	83.3	65.5	87.2	90.0	82.6	90.9	67.8	82.5
OPT-13B	SubZero+/SGD	95.4	84.8	84.2	69.6	87.2	92.0	82.0	90.7	67.5	83.7
	SubZero+/亚当	93.3	84.1	85.0	69.4	87.6	91.0	82.2	90.8	66.6	83.3
	零样本 FO (亚当)	58.8	59.6	59.0	55.0	46.9	80.0	81.2	46.2	14.6	55.7
		95.1	83.4	80.6	68.0	78.9	88.0	81.8	87.0	34.7	77.5
	MeZO	91.3	67.5	67.1	59.1	61.5	87.0	81.4	83.6	30.6	69.9
OPT-13B	绝地零	90.8	66.4	68.3	58.3	66.8	88.0	81.2	83.6	29.2	70.3
	SubZero+/SGD	92.8	70.8	74.4	60.7	69.5	88.0	81.2	85.3	30.6	72.5
	SubZero+/亚当	93.5	74.0	75.2	63.2	64.7	89.0	81.7	85.2	30.6	73.0

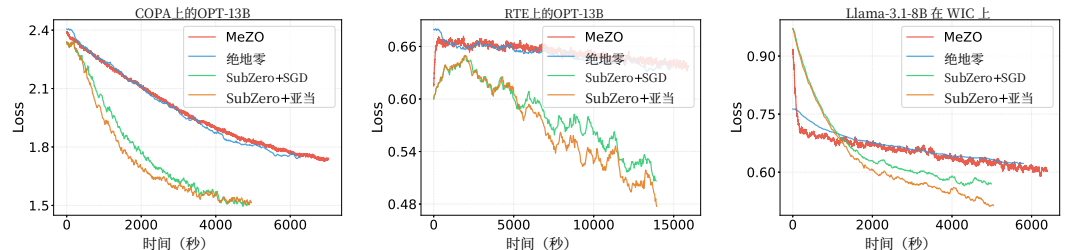


图4: 表1中选定损失曲线的可视化。

5.2.2 LoRA方案下的自回归语言模型

我们进一步在 OPT-13B、LLaMA3.1-8B 和 OPT-30B 上评估了 LoRA 方案下的 SubZero+ ($r = 8, \alpha = 16$)。LoRA 为注意力投影学习低秩适应矩阵，并在实践中被广泛使用。我们比较了每种 ZO 方法在超参数搜索中的最佳结果。结果如表 2 所示。如表 2 所示，SubZero+ 在 LoRA 方案下在两种模型规模上都优于 MeZO 和 SubZero。在 OPT-13B 上，SubZero+ 达到 70.7% 的平均准确率，比 MeZO (68.7%) 高出 +2.0%，比 SubZero (68.6%) 高出 +2.1%。在 LLaMA3.1-8B 上，SubZero+ 达到 83.4% 的平均准确率，比 MeZO (81.2%) 高出 +2.2%，比 SubZero (81.5%) 高出 +1.9%。对于更大的 OPT-30B，SubZero+ 进一步将平均准确率提升到 72.2%，超过 MeZO (70.1%) +2.1%，超过 SubZero (69.4%) +2.8%。这些结果与 FT 方案下获得的结果一致。

5.2.3 扩展至 QWEN2.5-32B

为了在一个大得多的模型上验证 SubZero+，我们还在 FT 方案下对 Qwen2.5-32B (Team, 2024) 进行了评估。我们将其与零样本基线、MeZO、带 LoRA 的 MeZO 以及 SubZero 进行比较。结果如表 3 所示。

在 Qwen2.5-32B 上，SubZero+ 在每个任务中都取得了最高分，平均准确率为 87.4%——比 SubZero (85.1%) 高出 +2.3%，比 MeZO (83.6%) 高出 +3.8%，比 MeZO-LoRA (76.7%) 高出 +10.7%，比零样本基线 (76.0%) 高出 +11.4%。在 SST-2 上的提升尤其明显 (比 SubZero 高出 +5.9%) 以及 WIC (比 SubZero 高出 +4.7%)。

Table 2: Main results on autoregressive LLMs under the LoRA scheme.

Model	Method	SST-2	RTE	BoolQ	WIC	MultiRC	COPA	ReCoRD	SQuAD	DROP	Avg.
	Task type	—— Classification ——				— Multi-choice —		— Generation —			
OPT-13B	MeZO	93.6	62.1	69.1	56.9	57.8	86.0	81.4	83.0	28.7	68.7
	SubZero	92.2	62.5	66.7	56.7	58.3	87.0	81.8	83.7	28.3	68.6
	SubZero+	93.0	71.5	73.9	60.2	60.5	86.0	80.4	82.5	28.6	70.7
LLaMA3.1-8B	MeZO	91.3	78.0	83.6	61.0	85.8	92.0	83.8	90.4	65.4	81.2
	SubZero	90.6	83.8	83.5	61.1	87.3	88.0	83.5	90.4	65.3	81.5
	SubZero+	93.3	84.8	84.1	69.0	87.6	89.0	84.3	91.1	66.9	83.4
OPT-30B	MeZO	92.6	68.2	63.6	60.0	65.4	86.0	80.5	83.6	30.6	70.1
	SubZero	93.0	65.3	61.4	58.2	64.0	85.0	81.7	83.2	32.5	69.4
	SubZero+	94.2	72.6	71.5	60.3	67.0	86.0	80.9	84.7	33.0	72.2

Table 3: Performance comparison on Qwen2.5-32B across tasks. All averages are computed over 5 tasks (SST-2, RTE, BoolQ, WIC, SQuAD). SubZero+ scores are oracle best values from task-specific optimal hyperparameter configurations.

Method	SST-2	RTE	BoolQ	WIC	SQuAD	Avg.
Zero-shot	61.5	87.0	84.3	61.8	85.5	76.0
MeZO	87.5	90.3	86.7	65.7	87.9	83.6
MeZO-LoRA	62.0	87.4	84.8	64.7	84.7	76.7
SubZero	89.1	90.3	87.7	68.8	89.6	85.1
SubZero+	93.4	91.3	88.5	73.5	90.3	87.4

5.2.4 LEARNING RATE ROBUSTNESS

A key claim of SubZero+ is that multi-query estimation dramatically improves learning rate robustness. We validate this by fine-tuning Qwen3-0.6B with LoRA on SST-2, sweeping learning rates across four orders of magnitude for both SubZero+ and MeZO. As shown in Figure 5(a), SubZero+ achieves the highest training accuracy of 86.5% at $\text{LR} = 2 \times 10^{-2}$, and retains above 79.0% accuracy across a $3 \times \text{LR}$ range ($10^{-2} \sim 3 \times 10^{-2}$). In comparison, MeZO reaches its best accuracy of 83.3% at $\text{LR} = 1 \times 10^{-4}$, but its performance window is considerably narrower. When LR is doubled to 2×10^{-4} , MeZO’s accuracy plummets by over 31 percentage points to 51.7%, whereas SubZero+ drops by only 0.5 percentage points when LR is increased by $1.5 \times$ to 3×10^{-2} .

To eliminate the scale difference in optimal LR_s, we normalize each method’s LR by its own optimum in Figure 5(b). The normalized view reveals that SubZero+ exhibits a broad, symmetric performance plateau around the optimal LR, with a standard deviation of merely 2.98% within the $[\text{opt}/2, 2 \times \text{opt}]$ interval. MeZO, on the other hand, shows highly asymmetric behavior: its accuracy degrades gracefully when LR is below the optimum but collapses abruptly when LR is even slightly above it. This suggests that SubZero+ is substantially more robust to learning rate selection in practical LoRA fine-tuning scenarios, where the exact optimal LR is often unknown a priori.

5.2.5 HAAR-CORRECTED QR-BASED SUBSPACE CONSTRUCTION

We conduct a comprehensive ablation study to quantify the contribution of Haar-corrected QR subspaces across tasks, optimizers, and evaluation metrics. The experimental setup is as follows: OPT-1.3B with full-parameter tuning, $K = 99$ queries, and a fixed total forward-pass budget of 30,000. We compare Haar-corrected QR subspaces (sign correction applied per Eq. (11)) against standard QR subspaces (raw output of `torch.linalg.qr`) under both subspace SGD and subspace Adam optimizers. Results are reported across 9 SuperGLUE tasks in Table 4.

As shown in Table 4, Haar-corrected subspaces improve average accuracy under both Adam (+0.7%, from 61.5% to 62.2%) and SGD (+0.3%, from 65.1% to 65.3%). The improvement is larger under Adam, consistent with the hypothesis that adaptive per-layer scaling benefits more from unbiased directional coverage. MultiRC shows the most consistent gain (+2.8% under Adam, +2.6% under SGD). RTE exhibits striking optimizer-dependent behavior: Haar correction provides +4.6% under

表2：在LoRA方案下自回归大型语言模型的主要结果。

模型	方法	SST-2	RTE	BoolQ	WIC	MultiRC	COPA	ReCoRD	SQuAD	DROP	平均
	任务类型	—— 分类 ——				— 多选 — — 生成 —					
OPT-13B	MeZO	93.6	62.1	69.1	56.9	57.8	86.0	81.4	83.0	28.7	68.7
	绝地零	92.2	62.5	66.7	56.7	58.3	87.0	81.8	83.7	28.3	68.6
	绝地零度+	93.0	71.5	73.9	60.2	60.5	86.0	80.4	82.5	28.6	70.7
LLaMA3.1-8B	MeZO	91.3	78.0	83.6	61.0	85.8	92.0	83.8	90.4	65.4	81.2
	绝地零	90.6	83.8	83.5	61.1	87.3	88.0	83.5	90.4	65.3	81.5
	绝地零度+	93.3	84.8	84.1	69.0	87.6	89.0	84.3	91.1	66.9	83.4
OPT-30B	MeZO	92.6	68.2	63.6	60.0	65.4	86.0	80.5	83.6	30.6	70.1
	绝地零	93.0	65.3	61.4	58.2	64.0	85.0	81.7	83.2	32.5	69.4
	绝地零度+	94.2	72.6	71.5	60.3	67.0	86.0	80.9	84.7	33.0	72.2

表3：Qwen2.5-32B 在各任务上的性能比较。所有平均值均基于 5 个任务计算（SST-2、RTE、BoolQ、WIC、SQuAD）。SubZero+ 分数为来自任务特定最优超参数配置的理论最佳值。

方法	SST-2	RTE	布尔问答	WIC	SQuAD	Avg.
零样本	61.5	87.0	84.3	61.8	85.5	76.0
MeZO	87.5	90.3	86.7	65.7	87.9	83.6
MeZO-LoRA	62.0	87.4	84.8	64.7	84.7	76.7
绝地零	89.1	90.3	87.7	68.8	89.6	85.1
绝地零度+	93.4	91.3	88.5	73.5	90.3	87.4

5.2.4 学习率稳健性

SubZero+ 的一个关键论点是多查询估计显著提高了学习率的鲁棒性。我们通过在SST-2上使用LoRA微调Qwen3-0.6B来验证这一点，同时对SubZero+ 和MeZO的学习率在四个数量级范围内进行扫描。如图5(a)所示，SubZero+ 在学习率 $= 2 \times 10^{-2}$ 时实现了最高训练准确率86.5%，并且在 $3 \times$ 学习率范围($10^{-2} \sim 3 \times 10^{-2}$)内保持在79.0%以上。相比之下，MeZO在学习率 $= 1 \times 10^{-4}$ 时达到其最佳准确率83.3%，但其性能窗口明显较窄。当学习率加倍到 2×10^{-4} 时，MeZO的准确率骤降超过31个百分点至51.7%，而SubZero+ 在学习率增加 $1.5 \times$ 至 3×10^{-2} 时仅下降0.5个百分点。

为了消除最佳学习率（LR）之间的尺度差异，我们在图5(b)中将每种方法的LR归一化为其自身的最优值。归一化后的视图显示，SubZero+ 在最优LR附近表现出宽广且对称的性能平台，其在 $[\text{opt}/2, 2 \times \text{opt}]$ 区间内的标准差仅为2.98%。另一方面，MeZO表现出高度不对称的行为：当LR低于最优值时，其准确率会平缓下降，但当LR略高于最优值时，准确率会突然崩溃。这表明在实际的LoRA微调场景中，SubZero+ 对学习率选择的鲁棒性显著更强，而在这些场景中，精确的最优LR通常是事先未知的。

5.2.5 基于QR的HAAR校正子空间构建

我们进行了一项全面的消融研究，以量化在不同任务、优化器和评估指标中 Haar 校正 QR 子空间的贡献。实验设置如下：OPT-1.3B 全参数微调， $K = 99$ 查询，以及固定的总前向计算预算为 30,000。我们将 Haar 校正 QR 子空间（按照公式 (11) 应用符号校正）与标准 QR 子空间（`torch.linalg.qr` 的原始输出）在子空间 SGD 和子空间 Adam 优化器下进行比较。结果在表 4 中展示了 9 个 SuperGLUE 任务的表现。

如表 4 所示，经过 Haar 校正的子空间在 Adam（+0.7%，从 61.5% 提高到 62.2%）和 SGD（+0.3%，从 65.1% 提高到 65.3%）下都能提高平均准确率。在 Adam 下的提升更大，这与假设相一致，即自适应每层缩放更受益于无偏方向覆盖。MultiRC 显示最稳定的提升（在 Adam 下为 +2.8%，在 SGD 下为 +2.6%）。RTE 表现出明显的优化器依赖行为：Haar 校正提供了 +4.6% 的提升

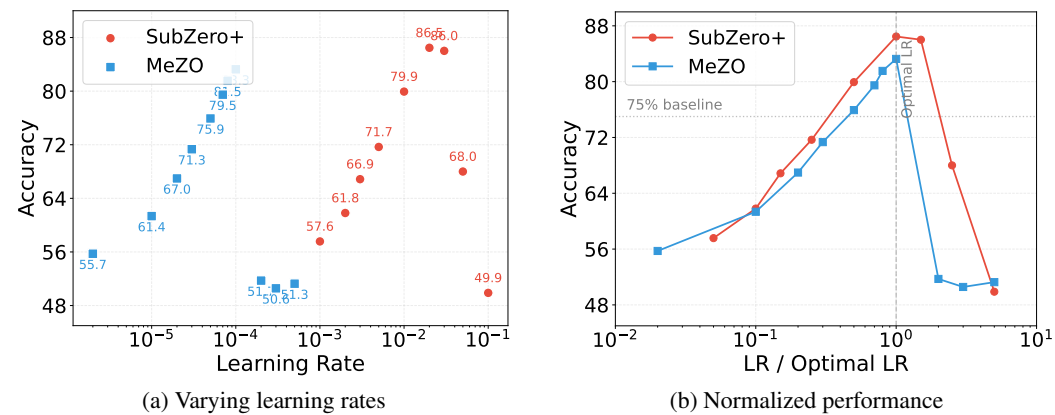


Figure 5: Learning rate robustness of SubZero+ and MeZO for LoRA of Qwen3-0.6B on SST-2. (a) Training accuracy under varying learning rates. SubZero+ maintains high accuracy across a wide LR range, while MeZO degrades sharply beyond its optimum. (b) Normalized view with each LR scaled by its optimal value. SubZero+ exhibits a symmetric plateau, while MeZO collapses when LR exceeds the optimum.

Table 4: Comparison of Haar-corrected and standard QR-based subspaces. Results on OPT-1.3B across 9 SuperGLUE tasks with both Adam and SGD subspace optimizers. “Haar” denotes Haar-corrected QR subspaces; “No Haar” denotes raw QR from `torch.linalg.qr` without sign correction. All experiments use subspace rank $r = 64$, $K = 99$ queries, and FT scheme. Best results between Haar and No Haar for each optimizer are bolded.

Task	Adam			SGD		
	Haar	No Haar	Δ	Haar	No Haar	Δ
BoolQ	65.6	68.2	-2.6	66.4	66.4	0.0
COPA	77.0	75.0	+2.0	75.0	74.0	+1.0
DROP	23.8	24.3	-0.4	25.4	24.2	+1.2
MultiRC	56.0	53.2	+2.8	61.7	59.1	+2.6
RTE	65.2	60.6	+4.6	60.3	63.5	-3.2
ReCoRD	57.2	61.2	-4.0	71.9	72.2	-0.3
SQuAD	67.2	65.9	+1.3	77.2	76.5	+0.7
SST-2	91.0	91.4	-0.4	90.0	91.3	-1.2
WIC	56.6	54.0	+2.6	60.0	58.5	+1.6
Avg.	62.2	61.5	+0.7	65.3	65.1	+0.3

Adam but -3.2% under SGD, suggesting that RTE’s loss landscape benefits from Haar-uniform exploration only when combined with adaptive step sizes. On SST-2 and ReCoRD, both optimizers show small differences ($< 1.3\%$), indicating low sensitivity to directional bias. We note that SGD achieves higher absolute accuracy than Adam on average, attributable to different learning rate search spaces. SGD with Haar correction remains a strong baseline for accuracy maximization, while subspace Adam is preferred for robustness and convergence speed.

5.3 QUERY COUNT ABLATION

Multi-query estimation is a central innovation of SubZero+, with the number of queries K controlling the trade-off between gradient estimate variance ($\propto 1/K$) and per-step computational cost ($\propto K$). We conduct a systematic ablation of K across three tasks with distinct characteristics: SST-2 (binary sentiment classification), RTE (binary entailment), and MultiRC (multi-choice reading comprehension).

Experimental setup. All experiments use OPT-1.3B with SubZero+ (Haar-corrected QR subspaces, $r = 24$, Z-space Adam), full-parameter tuning, and a fixed total forward-pass budget of 30,000. This means that as K increases, the number of training steps decreases proportionally (steps =

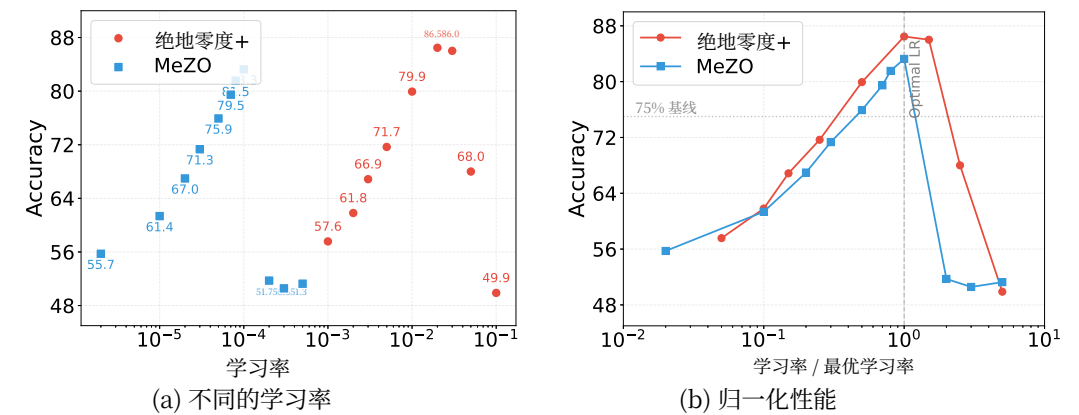


图5: SubZero+ 和MeZO在SST-2上对Qwen3-0.6B进行LoRA时的学习率稳健性。(a) 在不同学习率下的训练准确率。SubZero+ 在较宽的学习率范围内保持较高准确率，而MeZO在超过其最佳值时准确率急剧下降。(b) 将每个学习率按其最优值缩放后的归一化视图。SubZero+ 表现出对称的高原，而MeZO在学习率超过最优值时崩溃。

表 4: Haar 校正与标准基于 QR 的子空间比较。在 OPT-1.3B 上进行的 9 个 SuperGLUE 任务的结果，使用 Adam 和 SGD 子空间优化器。“Haar”表示 Haar 校正的 QR 子空间；“No Haar”表示来自 `torch.linalg.qr` 的原始 QR（未进行符号校正）。所有实验均使用子空间秩 $r = 64$, $K = 99$ 查询和 FT 方案。每个优化器下 Haar 与 No Haar 的最佳结果以粗体显示。

Task	Adam			SGD		
	Haar	无 Haar	Δ	Haar	无哈爾	Δ
布尔问答	65.6	68.2	-2.6	66.4	66.4	0.0
COPA	77.0	75.0	+2.0	75.0	74.0	+1.0
DROP	23.8	24.3	-0.4	25.4	24.2	+1.2
多重RC	56.0	53.2	+2.8	61.7	59.1	+2.6
RTE	65.2	60.6	+4.6	60.3	63.5	-3.2
记录	57.2	61.2	-4.0	71.9	72.2	-0.3
SQuAD	67.2	65.9	+1.3	77.2	76.5	+0.7
SST-2	91.0	91.4	-0.4	90.0	91.3	-1.2
WIC	56.6	54.0	+2.6	60.0	58.5	+1.6
Avg.	62.2	61.5	+0.7	65.3	65.1	+0.3

Adam 但在 SGD 下低 -3.2% ，这表明 RTE 的损失景观只有在结合自适应步长时，才会从 Haar 均匀探索中受益。在 SST-2 和 ReCoRD 上，两种优化器的差异都很小 ($< 1.3\%$)，表明对方向偏差的敏感性较低。我们注意到，SGD 的平均绝对准确率高于 Adam，这归因于不同的学习率搜索空间。带 Haar 校正的 SGD 仍然是最大化准确率的强基线，而子空间 Adam 更适合用于稳健性和收敛速度。

5.3 查询消融

多查询估计是 SubZero+ 的核心创新，查询数量 K 控制梯度估计方差 ($\propto 1/K$) 与每步计算成本 ($\propto K$) 之间的权衡。我们在三项具有不同特性的任务上系统地进行 K 的消融实验：SST-2（二元情感分类）、RTE（二元蕴含）和 MultiRC（多选阅读理解）。

实验设置。所有实验使用 OPT-1.3B 与 SubZero+（Haar 校正 QR 子空间， $r = 24$, Z 空间 Adam）、全参数调优，以及固定的总前向传播预算 30,000。这意味着随着 K 增加，训练步骤数按比例减少（步骤 =

Table 5: Ablation study on query count K (denoted q) across tasks. Results on OPT-1.3B with SubZero+ (Haar-corrected QR subspaces, $r = 24$, Z-space Adam) across SST-2, RTE, and MultiRC. Total forward-pass budget is fixed at 30,000 for all K ; larger K uses proportionally fewer steps. All experiments use full-parameter tuning. The best accuracy for each task is bolded, and the best overall K for the average is bolded in the Avg. column.

K (Queries)	SST-2	RTE	MultiRC	Avg.
1	50.9	47.7	55.5	51.4
5	52.6	52.7	55.5	53.6
10	56.7	49.8	52.3	52.9
20	69.8	57.0	59.8	62.2
30	88.3	62.8	57.6	69.6
50	90.9	63.9	60.0	71.6
70	91.5	58.1	58.2	69.3
80	91.6	60.3	59.3	70.4
100	92.2	57.0	59.6	69.6
150	89.9	61.4	58.7	70.0
200	89.7	61.0	61.1	70.6
Δ ($K=100$ vs. $K=1$)	+41.3	+9.3	+4.1	+18.2

Table 6: Peak GPU memory on OPT-13B (GB, batch size 16, fp16). Under FT, SubZero+ incurs $\sim 1\%$ overhead over MeZO due to per-layer subspace buffers. Under LoRA, all three methods consume identical memory because the trainable LoRA parameters dominate.

Task	FT				LoRA			
	MeZO	SubZero	SubZero+	$\Delta\%$	MeZO	SubZero	SubZero+	$\Delta\%$
RTE	30.4	30.5	30.7	+1.1	30.4	30.4	30.4	+0.0
BoolQ	34.1	34.2	34.5	+1.0	34.1	34.1	34.1	0.0
SQuAD	30.8	30.9	31.1	+1.0	30.8	30.8	30.8	+0.0
DROP	50.4	50.5	50.8	+0.6	50.4	50.4	50.4	0.0

$\lfloor 30,000/K \rfloor$). Learning rates are tuned per K via grid search (see Appendix A.2 for details). Results are presented in Table 5.

Table 5 reveals strikingly different K -accuracy profiles across tasks. SST-2 benefits dramatically from large K : accuracy rises from 50.9% at $K = 1$ (barely above random) to 92.2% at $K = 100$, a gain of +41.3 percentage points, with a critical jump at $K = 30$ (69.8% $\rightarrow$ 88.3%). RTE peaks at $K = 50$ (63.9%), with performance degrading for $K > 50$ as the reduced step count limits convergence. MultiRC is the least sensitive to K (varying only 52.3%–61.1%), suggesting that this task is bottlenecked by factors other than gradient estimation quality. For all three tasks, $K \leq 10$ yields accuracy within 5–7 points of random guessing, confirming that single-query estimates are too noisy for effective optimization.

5.4 MEMORY ANALYSIS

We measure the peak GPU memory usage of MeZO, SubZero, and SubZero+ using OPT-13B under both FT and LoRA schemes across four representative tasks. Results are presented in Table 6.

6 CONCLUSION

We have presented SubZero+, a memory-efficient ZO fine-tuning framework that integrates multi-query subspace gradient estimation, low-rank subspace Adam, and QR-based subspace construction with sign correction. Experiments across models (1.3B–32B), tasks, and FT/LoRA schemes demonstrate that SubZero+ consistently outperforms existing ZO baselines while maintaining inference-level memory, with substantially improved learning rate robustness and convergence speed.

Limitations and Future Work. The fundamental trade-off of SubZero+ is that its random subspaces are data-independent. While this ensures memory efficiency, it may also discard important gradient

预印本。

表 5: 关于各任务中查询次数 K (记作 q) 的消融研究。在 SST-2、RTE 和 MultiRC 上, 使用 SubZero+ (Haar 校正的 QR 子空间, $r = 24$, Z 空间 Adam) 的 OPT-1.3B 的结果。总前向传递预算在所有 K 中固定为 30,000; 更大的 K 使用的步骤相应更少。所有实验都使用全参数调优。每个任务的最佳准确率用粗体表示, 平均值列中整体最佳 K 也用粗体表示。

K (查询)	SST-2	RTE	多重RC	Avg.
1	50.9	47.7	55.5	51.4
5	52.6	52.7	55.5	53.6
10	56.7	49.8	52.3	52.9
20	69.8	57.0	59.8	62.2
30	88.3	62.8	57.6	69.6
50	90.9	63.9	60.0	71.6
70	91.5	58.1	58.2	69.3
80	91.6	60.3	59.3	70.4
100	92.2	57.0	59.6	69.6
150	89.9	61.4	58.7	70.0
200	89.7	61.0	61.1	70.6
Δ ($K=100$ 对 $K=1$)	+41.3	+9.3	+4.1	+18.2

表6: OPT-13B的峰值GPU内存 (GB, 批量大小16, fp16)。在FT下, 由于每层子空间缓冲区的存在, SubZero+ 相比MeZO会产生 $\sim 1\%$ 的开销。在LoRA下, 三种方法消耗的内存相同, 因为可训练的LoRA参数占主导地位。

Task	FT				LoRA			
	MeZO	绝地零	绝地零度+	$\Delta\%$	MeZO	绝地零	绝地零度+	$\Delta\%$
RTE	30.4	30.5	30.7	+1.1	30.4	30.4	30.4	+0.0
布尔问答	34.1	34.2	34.5	+1.0	34.1	34.1	34.1	0.0
SQuAD	30.8	30.9	31.1	+1.0	30.8	30.8	30.8	+0.0
DROP	50.4	50.5	50.8	+0.6	50.4	50.4	50.4	0.0

$\lfloor 30,000/K \rfloor$ 。学习率通过网格搜索针对 K 进行调优 (详情见附录 A.2)。结果展示在表 5 中。

表5显示了各任务之间截然不同的 K -准确率曲线。SST-2显著受益于较大的 K : 准确率从 $K = 1$ 的 50.9% (仅略高于随机) 上升到 $K = 100$ 的 92.2%, 提升了 +41.3 个百分点, 其中在 $K = 30$ 出现了关键跳跃 (69.8% $\rightarrow$ 88.3%)。RTE在 $K = 50$ 达到峰值 (63.9%), 随着 $K > 50$ 的减少步数限制了收敛, 性能有所下降。MultiRC对 K 的敏感度最小 (仅变化在 52.3%–61.1%之间), 这表明该任务的瓶颈在于梯度估计质量之外的因素。对于所有三个任务, $K \leq 10$ 的准确率均在随机猜测的 5–7 个百分点内, 证实单次查询估计对于有效优化来说噪声过大。

5.4 内存分析

我们在四个代表性任务中, 使用 OPT-13B 在 FT 和 LoRA 方案下, 测量了 MeZO、SubZero 和 SubZero+ 的峰值 GPU 内存使用情况。结果如表 6 所示。

6 结论

我们提出了 SubZero+, 一个内存高效的零阶 (ZO) 微调框架, 它整合了多查询子空间梯度估计、低秩子空间 Adam 以及基于 QR 的子空间构建与符号校正。在跨模型 (1.3B–32B)、任务以及 FT/LoRA 方案的实验中, 结果显示 SubZero+ 在保持推理级别内存的同时, 始终优于现有的 ZO 基线, 并显著提升了学习率鲁棒性和收敛速度。

局限性与未来工作。SubZero+ 的基本权衡在于其随机子空间是与数据无关的。虽然这保证了内存效率, 但它也可能会丢失重要的梯度

Preprint.

directions. In addition, the subspace is regenerated from scratch every F steps rather than being adapted based on prior optimization history. Incorporating data-dependent subspace construction and adaptive subspace updating are promising directions for future work.

Broader Impact. SubZero+ lowers the barrier to LLM customization by reducing memory cost and hyperparameter sensitivity. We see no specific negative societal impacts beyond those generally associated with LLM fine-tuning.

REFERENCES

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. GPT-4 technical report. *arXiv:2303.08774*, 2023.

Shun-ichi Amari. Backpropagation and stochastic gradient descent method. *Neurocomputing*, 5(4-5):185–196, 1993.

Yiming Chen, Yuan Zhang, Liyuan Cao, Kun Yuan, and Zaiwen Wen. Enhancing zeroth-order fine-tuning for language models with low-rank structures. In *Proceedings of the International Conference on Learning Representations*, 2025.

Sizhe Dang, Yanjun Zhao, Xiaodong Zheng, Guang Dai, Ivor Tsang, and Haishan Ye. FZOO: Fast zeroth-order optimizer for fine-tuning large language models towards adam-scale speed. In *International Conference on Learning Representations*, volume 2026, pp. 113760–113792, 2026.

John C Duchi, Michael I Jordan, Martin J Wainwright, and Andre Wibisono. Optimal rates for zero-order convex optimization: The power of two function evaluations. *IEEE Transactions on Information Theory*, 61(5): 2788–2806, 2015.

Saeed Ghadimi and Guanghui Lan. Stochastic first-and zeroth-order methods for nonconvex stochastic programming. *SIAM Journal on Optimization*, 23(4):2341–2368, 2013.

Athanasios Glentis, Dawei Li, Chung-Yiu Yau, and Mingyi Hong. Revisiting the Adam-SGD gap in LLM pre-training: The role of large effective learning rates. *arXiv:2605.17787*, 2026.

Wentao Guo, Jikai Long, Yimeng Zeng, Zirui Liu, Xinyu Yang, Yide Ran, Jacob R Gardner, Osbert Bastani, Christopher De Sa, Xiaodong Yu, et al. Zeroth-order fine-tuning of LLMs with extreme sparsity. *arXiv:2406.02913*, 2024.

Wentao Guo, Jikai Long, Yimeng Zeng, Zirui Liu, Xinyu Yang, Yide Ran, Jacob Gardner, Osbert Bastani, Christopher De Sa, Xiaodong Yu, et al. Zeroth-order fine-tuning of LLMs with transferable static sparsity. In *Proceedings of the International Conference on Learning Representations*, volume 2025, pp. 14814–14854, 2025.

Yongchang Hao, Yanshuai Cao, and Lili Mou. FLoRA: Low-rank adapters are secretly gradient compressors. In *Proceedings of the International Conference on Machine Learning*, 2024.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *Proceedings of the International Conference on Learning Representations*, 2022.

Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *Proceedings of the International Conference on Learning Representations*, 2015.

Yicheng Lang, Changsheng Wang, Yihua Zhang, Mingyi Hong, Zheng Zhang, Wotao Yin, and Sijia Liu. Powering up zeroth-order training via subspace gradient orthogonalization. *arXiv:2602.17155*, 2026.

Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pp. 3045–3059, 2021.

Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics and the International Joint Conference on Natural Language Processing*, pp. 4582–4597, 2021.

Yukang Lin et al. The multi-query paradox in zeroth-order optimization. *arXiv:2509.15552*, 2025.

预印本。

方向。此外，该子空间每隔 F 步从零重新生成，而不是基于先前的优化历史进行调整。将数据依赖的子空间构建和自适应子空间更新纳入考虑是未来工作的有前景方向。

更广泛的影响。SubZero+ 通过降低内存成本和超参数敏感性，降低了大语言模型定制的门槛。除了通常与大语言模型微调相关的影响之外，我们没有看到具体的负面社会影响。

参考文献

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat 等. GPT-4 技术报告。arXiv:2303.08774, 2023。

天利 俊一。反向传播和随机梯度下降方法。神经计算， 5 196， 1993 4 5 185(-): - 。

陈一鸣, 张源, 曹丽媛, 袁坤, 文在文. 利用低秩结构增强语言模型的零阶微调. 载于《国际学习表征会议论文集》, 2025.

党思哲, 赵彦军, 那晓博, 戴浩, 王德岭, 李海山. 面向单级别速度的快速零阶优化器. 发表于《国际学习表征会议》, 2026 年, 第 2026 卷, 页 113760–113792。

约翰·C·杜奇, 迈克尔·I·乔丹, 马丁·J·韦恩赖特, 安德烈·韦比索尼. 零阶凸优化的最优速率: 两次函数评估的充分性. 《统计科学理论汇刊》, 61(5): 2788–2806, 2015。

赛义德·穆迪和广辉·兰. 随机规划的随机一阶和零阶方法。《SIAM优化杂志》, 23(4):2341–2368, 2013。

Athanasios Glentis, 李大为, 邱仲耀, 和洪明义. 重新审视LLM预训练中的Adam-SGD差距：大有效学习率的作用。arXiv:2605.17787, 2026。

郭文涛、龙继凯、曾一萌、刘子睿、杨心宇、冉一德、雅各布·R·加德纳、奥斯伯特·巴斯塔尼、克里斯托弗·德萨、于晓东 等。极度稀疏的大型语言模型零阶微调。arXiv:2406.02913,2024。

郭文涛, 龙际凯, 曾一萌, 刘子睿, 杨心宇, 冉一德, 雅各布·加德纳, 奥斯伯特·巴斯塔尼, 克里斯托弗·德萨, 余晓东 等人。具有可转移静态稀疏性的 LLMs 零阶微调。载于《国际学习表征会议论文集》, 卷 2025, 第 14854 页, 2025, 14814 页–。

郝永昌, 曹彦帅, 牟莉莉. FLoRA：低秩适配器实际上是隐秘的梯度压缩器。载于《国际机器学习会议论文集》, 2024年。

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, 和 Weizhu Chen. LoRA：大语言模型的低秩适应。在国际学习表示大会论文集， 2022年。

迪德里克·P·金马和吉米·巴. Adam：一种随机优化方法。载于2015年国际学习表征会议论文集。

朗宜诚, 王长盛, 张奕华, 洪明义, 张铮, 尹沃涛, 刘思佳. 通过子空间梯度正交化增强零阶训练. arXiv:2602.17155, 2026.

Brian Lester, Rami Al-Rfou, 和 Noah Constant. 参数高效提示调优的规模效应。发表于《自然语言处理实证方法会议论文集》, 第 3045–3059 页, 2021 年。

向丽莎·李和 Percy Liang. 前缀调优：优化连续提示以用于生成。载于《自然语言处理实证方法会议论文集》, 第 4582–4597 页, 2021 年。

林玉康 等 . 零阶优化中的多查询悖论 . *arXiv:2509 . 15552*, 2025 .

A APPENDIX

A.1 IMPLEMENTATION DETAILS

For the subspace construction with sign correction, we draw random Gaussian matrices $\mathbf{\Omega}_U \in \mathbb{R}^{m_i \times r}$, $\mathbf{\Omega}_V \in \mathbb{R}^{n_i \times r}$, compute standard thin QR decomposition (PyTorch’s `torch.linalg.qr`), and apply the sign correction as described in §4.3 (multiplying each column of $\tilde{\mathbf{U}}_i, \tilde{\mathbf{V}}_i$ by the sign of the corresponding diagonal entry of $\tilde{\mathbf{R}}_U, \tilde{\mathbf{R}}_V$).

SubZero+ inherits the random seed trick and per-layer parameter update strategy from SubZero (Yu et al., 2025) for memory-efficient perturbation regeneration. The subspace projection matrices $\mathbf{U}_i, \mathbf{V}_i$ are regenerated every F steps and cached in between. All Adam moment buffers $\mathbf{M}_i^t, \mathbf{V}_i^t$ are stored in $\mathbb{R}^{r \times r}$, contributing only $2r^2$ floats per layer. The multi-query perturbations are generated sequentially within each iteration, with each query’s perturbation regenerated from its random seed before the forward pass and then immediately subtracted after the loss computation, keeping peak memory at inference levels.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

Hyperparameter Configurations. All experiments are conducted on a single NVIDIA H200 (141GB) GPU, except for OPT-1.3B which is run on a single NVIDIA RTX 4090 (24GB) GPU. We use PyTorch 2.1.0 with CUDA 12.1.

For MeZO and SubZero, we follow the hyperparameter search grids from prior work (Malladi et al., 2023; Yu et al., 2025). For SubZero+, we additionally tune the subspace rank r , subspace update frequency F , queries K , and Adam parameters. For full-parameter tuning (FT), the subspace rank r is model-dependent: $r = 16$ for OPT-1.3B and $r = 128$ for all other models (OPT-13B, LLaMA3.1-8B), following the same scheme as SubZero. Tables 7 and 8 summarize the search grids for the FT and LoRA schemes, respectively. All ZO methods use batch size 16 and constant learning rate schedule. Learning rates and perturbation scales are selected via grid search on the validation set.

Table 7: Hyperparameter search grids for the FT scheme.

Method	Hyperparameter	Value
MeZO	batch size	16
	learning rate	{1e-7, 5e-7, 1e-6}
	ϵ	1e-3
	training steps	20,000
SubZero	batch size	16
	learning rate	{1e-7, 5e-7, 1e-6}
	ϵ	1e-3
	rank r	16 (OPT-1.3B) / 128 (others)
	subspace update freq. F	{500, 1000, 2000}
SubZero+	training steps	20,000
	batch size	16
	learning rate	{1e-3, 5e-3, 1e-2}
	ϵ	1e-3
	rank r	16 (OPT-1.3B) / 128 (others)
	subspace update freq. F	{50, 100}
	queries K	99
	training steps	400
	Adam β_1, β_2	0.9, 0.95

Forward-Pass Budget. MeZO and SubZero employ two-point (central difference) gradient estimation, consuming 2 forward passes per step, for a total forward-pass budget of $2 \times 20K = 40K$. SubZero+ uses $K + 1$ forward passes per step (K perturbed and one baseline). For fair comparison, we adjust SubZero+’s step count so that its total forward-pass budget matches the 40K budget of MeZO and SubZero. With the default $K = 99$, SubZero+ runs for 400 steps, consuming $99 + 1 = 100$ forward passes per step, matching the 40K budget. All methods use the same total forward-pass budget in all experiments unless otherwise noted.

预印本。

附录 A

A.1 实施细节

对于带符号校正的子空间构造，我们生成随机高斯矩阵 $\mathbf{\Omega}_U \in \mathbb{R}^{m_i \times r}$, $\mathbf{\Omega}_V \in \mathbb{R}^{n_i \times r}$ ，计算标准薄 QR 分解（PyTorch 的 `torch.linalg.qr`）， $\tilde{\mathbf{U}}_i, \tilde{\mathbf{V}}_i$ 并按照 § 4.3 的描述应用符号校正（将 $\mathbf{U}_i, \mathbf{V}_i$ 每一列乘以 $\mathbf{R}_U, \mathbf{R}_V$ 对应对角元的符号 $\tilde{}, \tilde{}$ ）。

SubZero+ 继承了 SubZero (Yu 等, 2025) 的随机种子技巧和每层参数更新策略，以实现内存高效的扰动再生成。子空间投影矩阵 $\mathbf{U}_i, \mathbf{V}_i$ 每 F 步重新生成，并在此期间进行缓存。所有 Adam 动量缓冲 $\mathbf{M}_i^t, \mathbf{V}_i^t$ 都存储在 $\mathbb{R}^{r \times r}$ 中，每层仅占用 $2r^2$ 个浮点数。多查询扰动在每次迭代中顺序生成，每个查询的扰动在前向传播前从其随机种子重新生成，然后在损失计算后立即减去，从而将峰值内存保持在推理阶段水平。

A.2 额外的实验细节

超参数配置。所有实验都在单个 NVIDIA H200 (141GB) GPU 上进行，OPT-1.3B 除外，它在单个 NVIDIA RTX 4090 (24GB) GPU 上运行。我们使用 PyTorch 2.1.0 和 CUDA 12.1。

对于 MeZO 和 SubZero，我们遵循先前工作的超参数搜索网格（Malladi 等, 2023; Yu 等, 2025）。对于 SubZero+，我们还调整子空间秩 r 、子空间更新频率 F 、查询 K 以及 Adam 参数。对于全参数调优 (FT)，子空间秩 r 取决于模型：OPT-1.3B 为 $r = 16$ ，其他所有模型（OPT-13B、LLaMA3.1-8B）为 $r = 128$ ，遵循与 SubZero 相同的方案。表 7 和表 8 分别总结了 FT 和 LoRA 方案的搜索网格。所有 ZO 方法使用批量大小为 16 的固定学习率调度。学习率和扰动规模通过在验证集上的网格搜索选择。

表7：FT方案的超参数搜索网格。

方法	超参数	价值
MeZO	批量大小	16
	学习率	{1e-7, 5e-7, 1e-6}
	ϵ	1e-3
	训练步骤	20,000
绝地零	批量大小	16
	学习率	{1e-7, 5e-7, 1e-6}
	ϵ	1e-3
	排名 r	16 (OPT-1.3B) / 128 (其他)
	子空间更新频率 F	{500, 1000, 2000}
绝地零度+	训练步骤	20,000
	批量大小	16
	学习率	{1e-3, 5e-3, 1e-2}
	ϵ	1e-3
	排名 r	16 (OPT-1.3B) / 128 (其他)
	子空间更新频率 F	{50, 100}
	查询 K	99
	训练步骤	400
	亚当 β_1, β_2	0.9, 0.95

前向传递预算。MeZO 和 SubZero 使用双点（中心差分）梯度估计，每步消耗 2 次前向传递，总前向传递预算为 $2 \times 20K = 40K$ 。SubZero+ 每步使用 $K + 1$ 次前向传递（ K 次扰动和一次基线）。为了公平比较，我们调整 SubZero+ 的步数，使其总前向传递预算与 MeZO 和 SubZero 的 40K 预算相匹配。使用默认 $K = 99$ ，SubZero+ 运行 400 步，每步消耗 $99 + 1 = 100$ 次前向传递，与 40 K 的预算相符。除非另有说明，所有方法在所有实验中使用相同的总前向传递预算。

Table 8: Hyperparameter search grids for the LoRA scheme. LoRA rank is 8 and $\alpha = 16$ for all methods.

Method	Hyperparameter	Value
MeZO	batch size	16
	learning rate	$\{1.5\text{e-}5, 3\text{e-}5, 5\text{e-}5\}$
	ϵ	$1\text{e-}3$
	training steps	20,000
SubZero	batch size	16
	learning rate	$\{1.5\text{e-}5, 3\text{e-}5, 5\text{e-}5\}$
	ϵ	$1\text{e-}3$
	rank r	$\{4, 8, 16\}$
	subspace update freq. F	$\{500, 1000, 2000\}$
SubZero+	training steps	20,000
	batch size	16
	learning rate	$\{5\text{e-}3, 1\text{e-}2, 5\text{e-}2\}$
	ϵ	$1\text{e-}3$
	rank r	$\{4, 8, 16\}$
	subspace update freq. F	$\{50, 100\}$
	queries K	99
	training steps	400
	Adam β_1, β_2	0.9, 0.95

A.3 EMPIRICAL VALIDATION OF HAAR-CORRECTED QR SUBSPACES

We empirically validate that `torch.linalg.qr` without sign correction fails to produce Haar-distributed matrices. We generate $m \times r$ matrices with i.i.d. $\mathcal{N}(0, 1)$ entries ($m = 50, r = 10$), compute QR decomposition, and extract the first column of Q with and without sign correction (Eq. (11)). As reference, we use normalized Gaussian vectors (uniform on S^{m-1}). Results are averaged over 2000 trials.

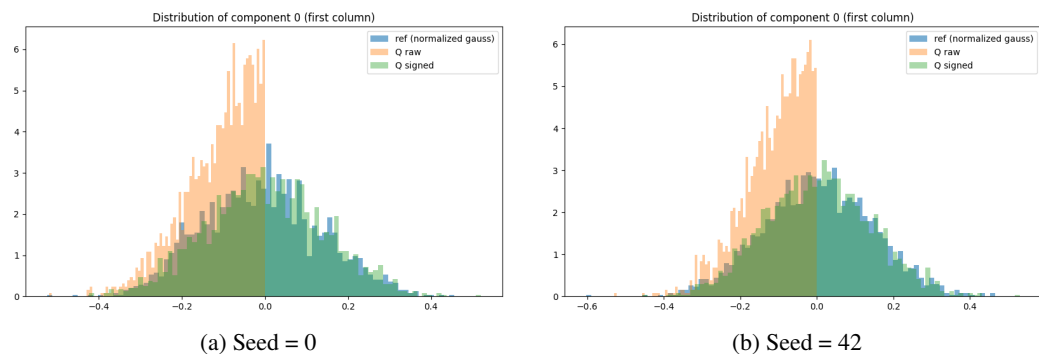


Figure 6: `torch.linalg.qr` without sign correction does not produce Haar-distributed Q . Histograms of a single coordinate of the first column of $Q \in \mathbb{R}^{50 \times 10}$ ($n = 2000$). Blue: reference (normalized Gaussian, uniform on S^{49}). Red: raw Q from `torch.linalg.qr`. Green: Q after sign correction. The raw Q deviates systematically from the Haar distribution, while the sign-corrected Q closely matches the reference.

Table 9: Two-sample KS test against the Haar reference. $D = \text{KS statistic}$ (lower is better). $p > 0.05$ indicates failure to reject the null of identical distributions.

Method	Seed	KS statistic D	p -value
Raw QR (<code>torch.linalg.qr</code>)	0	0.071	1.2×10^{-4}
Sign-corrected QR (Haar)	0	0.019	0.47
Raw QR (<code>torch.linalg.qr</code>)	42	0.068	3.1×10^{-4}
Sign-corrected QR (Haar)	42	0.021	0.38

Tab第8部分: LoRA 方案的超参数搜索网格。LoRA 等级为 8, $\alpha = 16$ 用于 a met箱式运

方法	超参数	价值
MeZO	批量大小	16
	学习率	$\{1.5\text{e-}5, 3\text{e-}5, 5\text{e-}5\}$
	ϵ	$1\text{e-}3$
	训练步骤	20,000
绝地零	批量大小	16
	学习率	$\{1.5\text{e-}5, 3\text{e-}5, 5\text{e-}5\}$
	ϵ	$1\text{e-}3$
	排名 r	$\{4, 8, 16\}$
	子空间更新频率 F	$\{500, 1000, 2000\}$
绝地零度+	训练步骤	20,000
	批量大小	16
	学习率	$\{5\text{e-}3, 1\text{e-}2, 5\text{e-}2\}$
	ϵ	$1\text{e-}3$
	排名 r	$\{4, 8, 16\}$
	子空间更新频率 F	$\{50, 100\}$
	查询 K	99
	训练步骤	400
	亚当 β_1, β_2	0.9, 0.95

A.3 HAAR校正QR子空间的实证验证

我们通过实验证实, 不进行符号校正的 `torch.linalg.qr` 无法生成 Haar 分布的矩阵。我们生成了 $m \times r$ 个条目服从独立同分布 $\mathcal{N}(0, 1)$ 的矩阵 ($m = 50, r = 10$), 计算 QR 分解, 并提取 Q 的第一列, 分别有无符号校正 (公式 (11))。作为参考, 我们使用归一化的高斯向量 (在 S^{m-1} 上均匀分布)。结果是 2000 次试验的平均值。

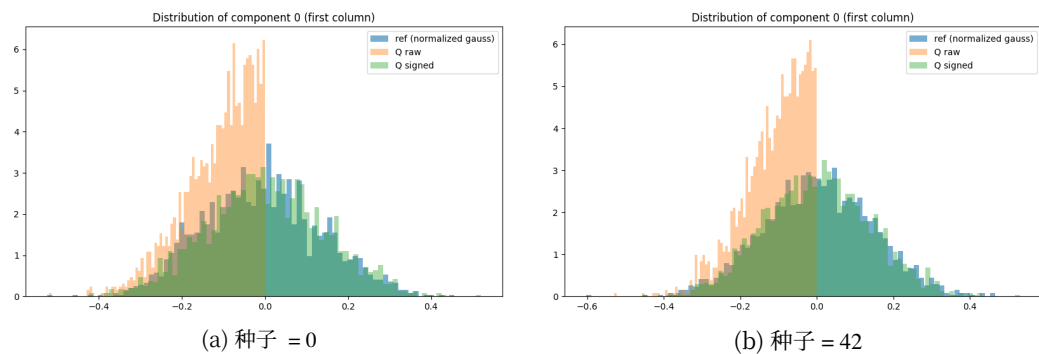


图6: `torch.linalg.qr` 未进行符号校正不会产生 Haar 分布的 Q 。 $Q \in \mathbb{R}^{50 \times 10}$ ($n = 2000$) 第一列单个坐标的直方图。蓝色: 参考 (归一化高斯, S^{49} 上的均匀分布)。红色: 来自 `torch.linalg.qr` 的原始 Q 。绿色: 符号校正后的 Q 。原始 Q 系统地偏离 Haar 分布, 而符号校正后的 Q 与参考分布高度吻合。

表9: 两样本K-S检验与Haar参考。 $D = \text{KS统计量}$ (越低越好)。 $p > 0.05$ 表示未能拒绝分布相同的原假设。

方法	Seed	KS 统计量 D	p -值
原始 QR (<code>torch.linalg.qr</code>)	0	0.071	1.2×10^{-4}
符号校正的二维码 (Haar)	0	0.019	0.47
原始 QR (<code>torch.linalg.qr</code>)	42	0.068	3.1×10^{-4}
符号校正的二维码 (Haar)	42	0.021	0.38

Figure 6 shows the empirical distribution of a single coordinate of the first column of Q . The sign-corrected Q (green) is virtually indistinguishable from the reference (blue), while the raw Q (red) is concentrated closer to zero with lighter tails. Table 9 confirms this quantitatively: the raw Q is strongly rejected by the KS test ($p \ll 0.001$), while the sign-corrected Q passes ($p > 0.05$). The same holds across different random seeds, CPU and GPU backends, and both standard and column-pivoted QR variants. The empirical variance of the raw Q 's first coordinate is 0.0177 (vs. theoretical 0.02), an 11.5% deficit confirming the broken rotational symmetry, while the sign-corrected Q achieves 0.0201, consistent with theory.

图6显示了 Q 第一列单个坐标的经验分布。符号校正后的 Q (绿色)几乎与参考(蓝色)无法区分，而原始 Q (红色)的分布更集中于零附近，尾部较轻。表9在数量上证实了这一点：原始 Q 在KS检验中被强烈拒绝($p \ll 0.001$)，而符号校正后的 Q 通过了检验($p > 0.05$)。这种情况在不同随机种子、CPU和GPU后端，以及标准和列主元QR变体中均成立。原始 Q 第一坐标的经验方差为0.0177（理论值为0.02），比理论值低11.5%，确认了旋转对称性被破坏，而符号校正后的 Q 达到0.0201，与理论一致。